Unit – 4

Supervised Learning – Classification, Regression

- Example of supervised learning,
- Classification model,
- Classification learning steps,
- Common classification algorithms,
- Example of regression,
- Common regression algorithms,.

Example of supervised learning

- In supervised learning, the labeled training data provide the basis for learning.
- According to the definition of machine learning, **this labeled training data is the experience or prior knowledge or belief.**
- It is called supervised learning because the process of learning from the **training data by a machine can be related to a teacher supervising the learning process of a student who is new to the subject**. Here, the teacher is the training data.
- Training data is the past information with known value of class field or 'label'.
- Hence, we say that **the 'training data is labeled' in the case of supervised learning** (refer Fig. 7.1).
- Different to this, there is **no labelled training data for unsupervised learning**.
- **Semi-supervised learning**, as depicted in Figure 7.1, **uses a small amount of labelled data along with unlabelled data for training.**

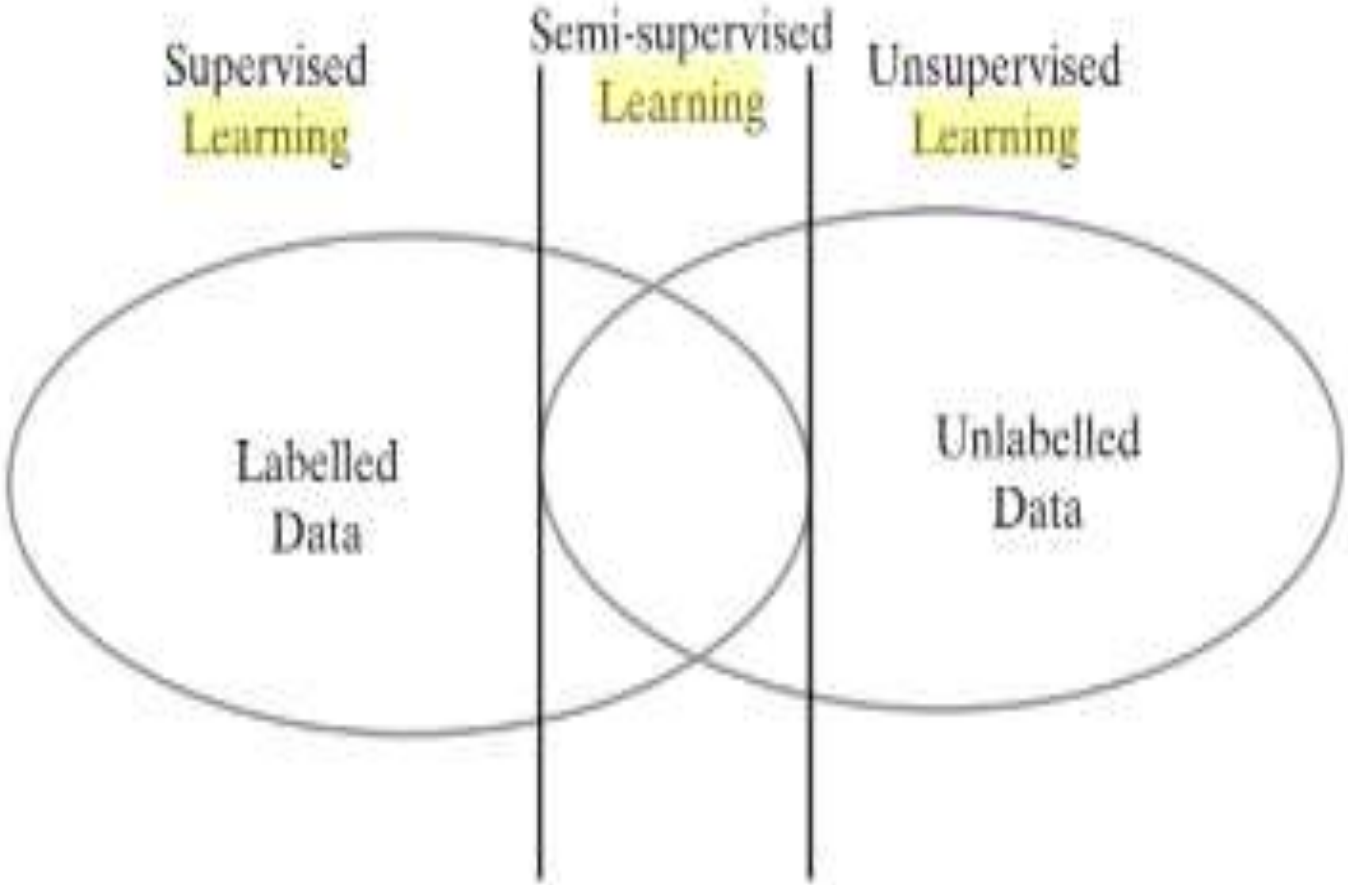


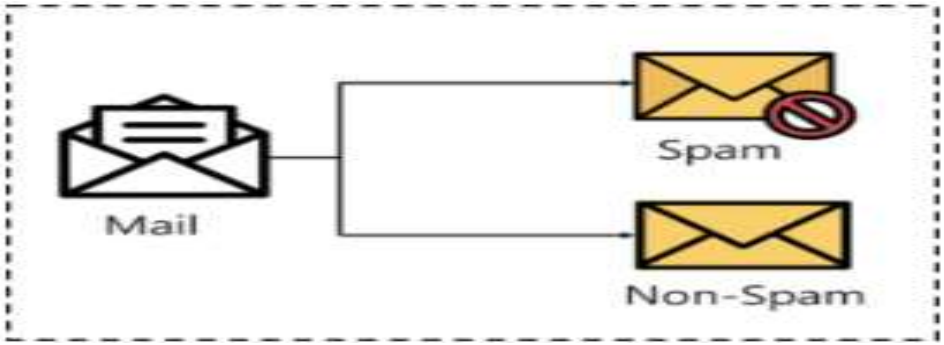
FIG. 7.1
Supervised learning vs. unsupervised learning

In a hospital, many patients are treated in the general wards. In comparison, the number of beds in the Intensive Care Unit (ICU) is much less. So, it is always a cause of worry for the hospital management that if the health condition of a number of patients in the general ward suddenly aggravates and they would have to be moved to ICU. Without previous planning and preparations, such a spike in demand becomes difficult for the hospital to manage. This problem can be addressed in a much better way if it is possible to predict which of the patients in the normal wards have a possibility of their health condition deteriorating and thus need to be moved to ICU.

This kind of prediction problem comes under the purview of supervised learning or, more specifically, under classification. The hospital already has all past patient records. The records of the patients whose health condition aggravated in the past and had to be moved to ICU can form the training data for this prediction problem. Test results of newly admitted patients are used to classify them as high-risk or low-risk patients.

What is Classification?

- Classification is a process of **categorizing a given set of data into classes, It can be performed on both structured or unstructured data.**
- **The process starts with predicting the class of given data points**. The classes are often referred to as target, label or categories.
- The classification predictive modeling is the task of approximating the mapping function from input variables to separate output variables.
- **The main goal is to identify which class/category the new data will fall into.**



- **Heart disease detection** can be identified as a classification problem, this is a binary classification since there can be only two classes i.e has heart disease or does not have heart disease. The classifier, in this case, needs training data to understand how the given input variables are related to the class. And once the classifier is trained accurately, it can be used to detect whether heart disease is there or not for a particular patient.

- Since classification is a type of supervised learning, even the targets are also provided with the input data.

Types Of Learners In Classification

- **Lazy Learners** – Lazy learners simply store the training data and **wait until a testing data appears**.
- The classification is done **using the most related data in the stored training data.**
- They have more predicting time compared to eager learners. **Eg – k-nearest neighbor, case-based reasoning.**

- **Eager Learners** – Eager learners construct a classification model based on the **given training data before getting data for predictions.**
- It must be able to commit to a single hypothesis that will work for the entire space.
- Due to this, **they take a lot of time in training and less time for a prediction**.
- **Eg – Decision Tree, Naive Bayes, Artificial Neural Networks.**

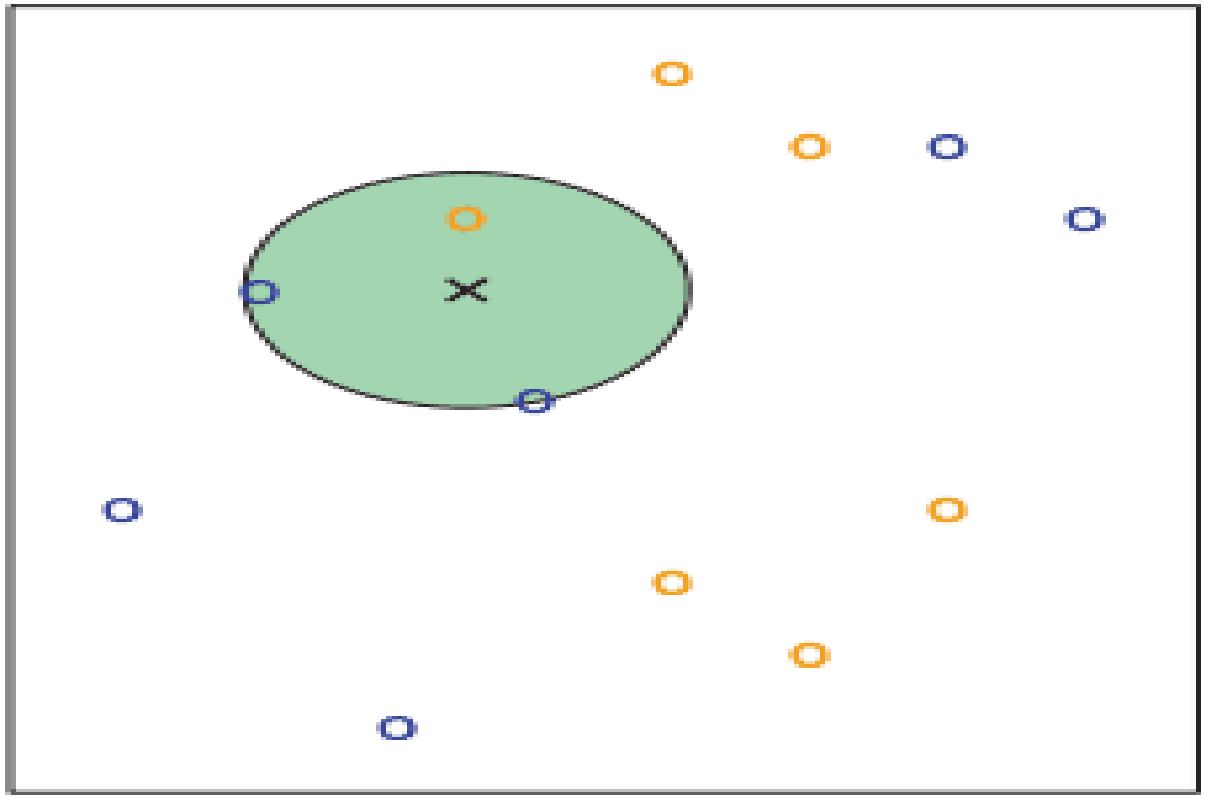
Common classification algorithms

- 1. k-Nearest Neighbour (kNN)
- 2. Decision Tree
- 3. Random Forest
- 4. Support Vector Machine (SVM)
- 5. Naïve Bayes classifier

K-Nearest Neighbor

- It is a lazy learning algorithm that **stores all instances corresponding to training data in n-dimensional space**.
- It is a **lazy learning algorithm** as it does not focus on constructing a general internal model, instead, it works on storing instances of training data.
- ***K nearest neighbors*** is a simple algorithm used for both classification and regression problems.
- It basically stores all available cases to classify the new cases by a majority vote of its k neighbors.
- The **case assigned to the class is most common amongst its K nearest neighbors measured by a distance function (Euclidean, Manhattan, Minkowski, and Hamming).**

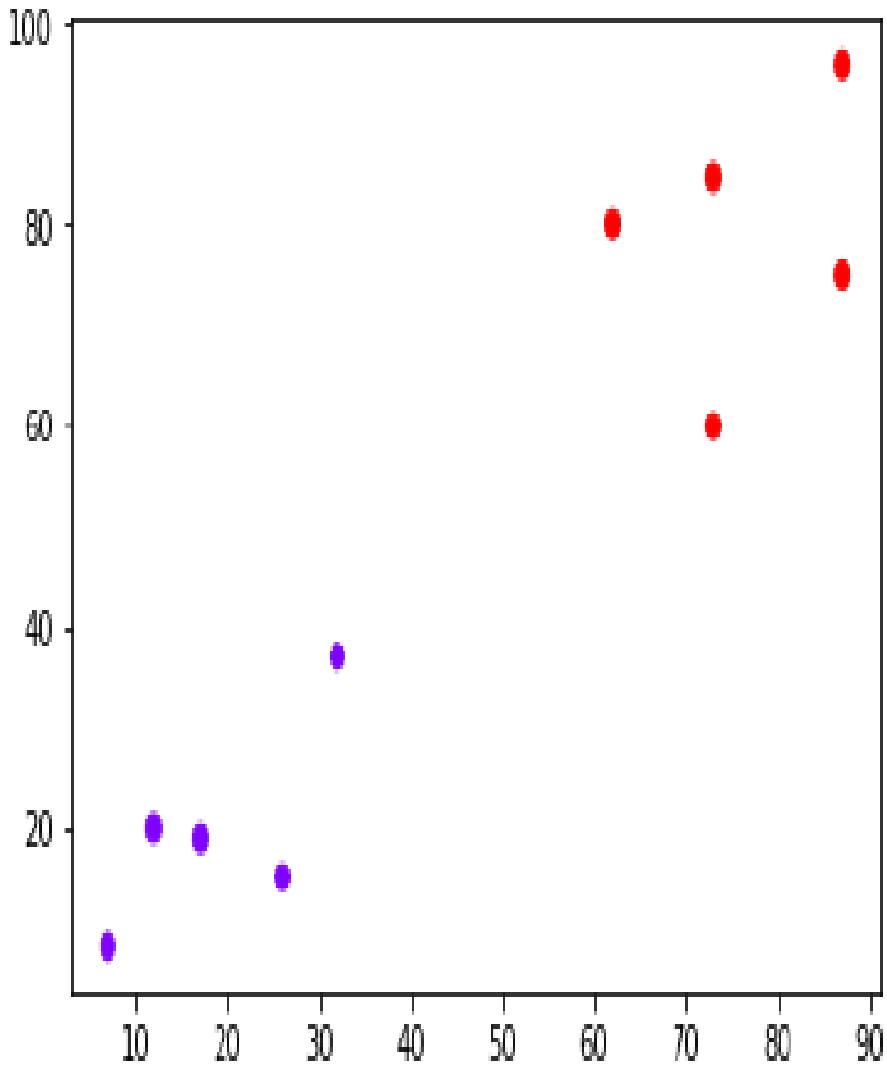
- **It first identifies the k points in the training data that are closest to the test value** and **calculates the distance between all those categories**.
- The test value will **belong to the category whose distance is the least**.



How kNN algorithm works?

- **Step 1** – For implementing any algorithm, **we need dataset**. So during the first step of KNN, we must load the training as well as test data.
- **Step 2** – Next, we need to **choose the value of K** i.e. the nearest data points. K can be any integer.
- **Step 3** – For each point in the test data do the following –
  - **3.1 – Calculate the distance between test data and each row of training data with the help of any of the method namely: Euclidean**, Manhattan or Hamming distance. **The most commonly used method to calculate distance is Euclidean.**
  - **3.2** – Now, based on the distance value, **sort them** in ascending order.
  - **3.3** – Next, it will choose the **top K rows** from the sorted array.
  - **3.4** – Now, it will **assign a class to the test** point based on most frequent class of these rows.
- **Step 4** – End

Example


100
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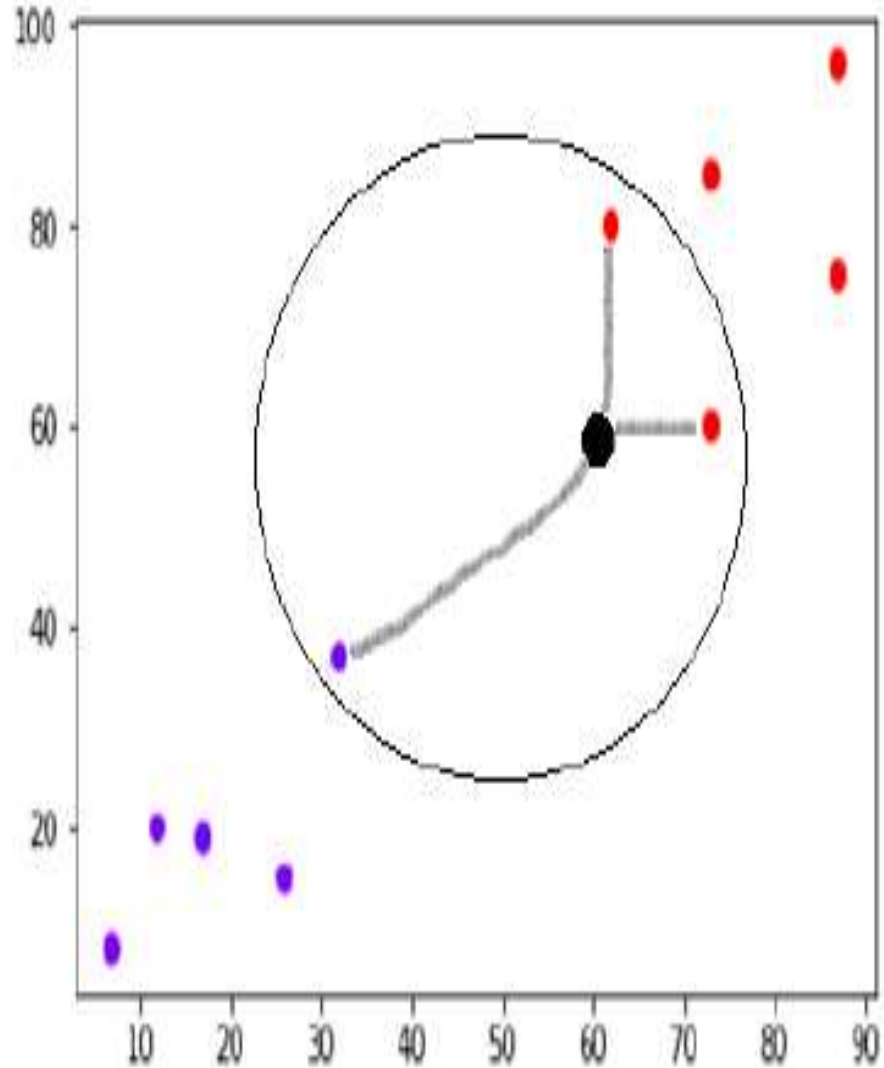
100
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70
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90

Example

Consider a dataset having two variables: height (cm) & weight (kg) and each point is classified as Normal or Underweight

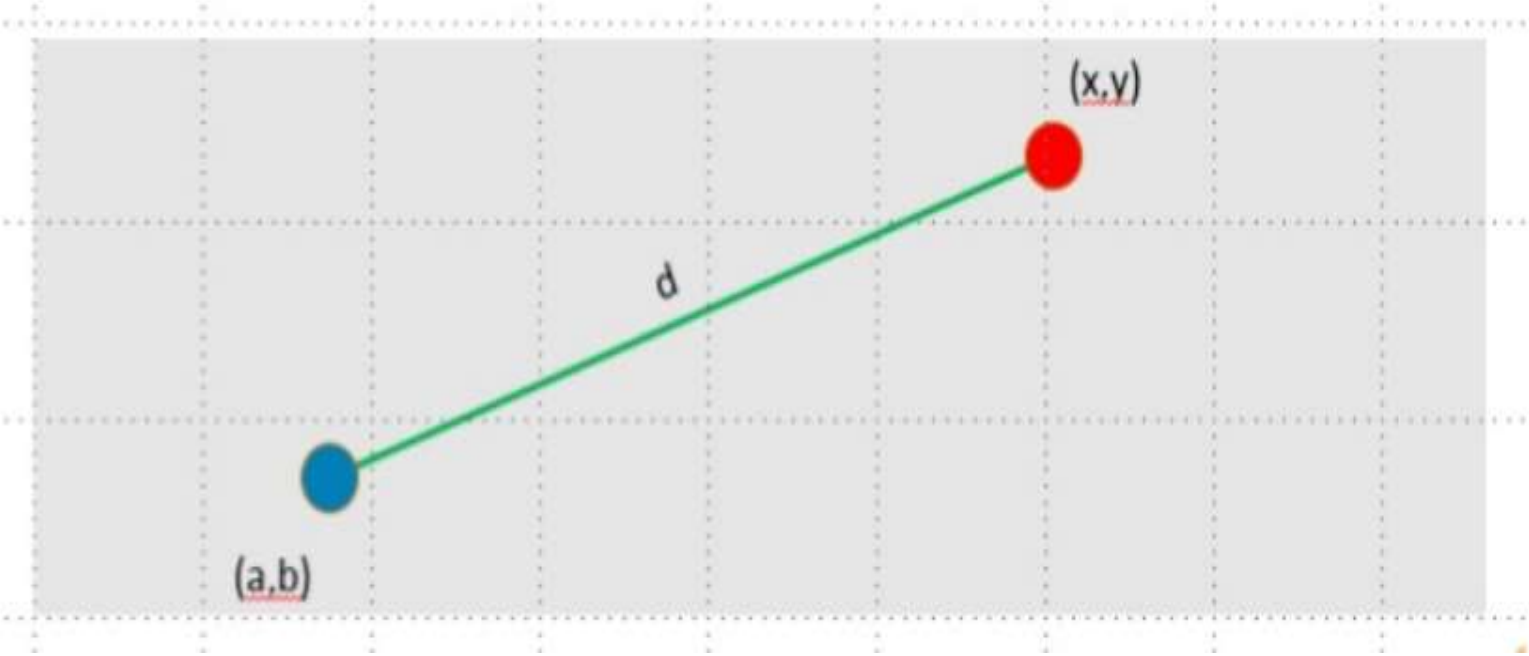
Weight(x2)	Height(y2)	Class
51	167	Underweight
62	182	Normal
69	176	Normal
64	173	Normal
65	172	Normal
56	174	Underweight
58	169	Normal
57	173	Normal
55	170	Normal

On the basis of the given data we have to classify the below set as Normal or Underweight using KNN

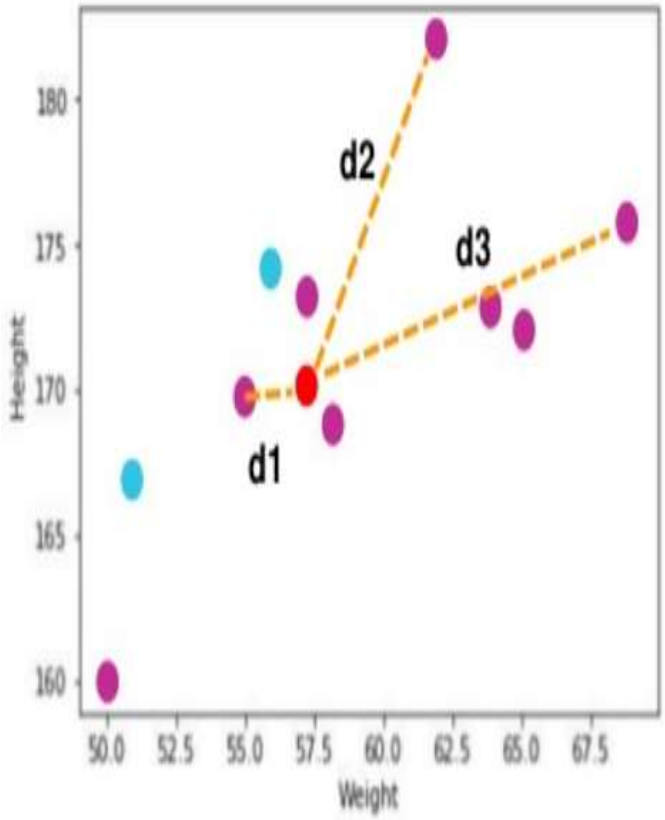
57 kg	170 cm	?

According to the **Euclidean distance** formula, the **distance** between two points in the plane with coordinates (x, y) and (a, b) is given by:

$$\text{dist}(d) = \sqrt{(x - a)^2 + (y - b)^2}$$



Let's calculate it to understand clearly:



dist(**d1**)= $\sqrt{(170-167)^2 + (57-51)^2} \sim= 6.7$

dist(**d2**)= $\sqrt{(170-182)^2 + (57-62)^2} \sim= 13$

dist(**d3**)= $\sqrt{(170-176)^2 + (57-69)^2} \sim= 13.4$

Similarly, we will calculate Euclidean distance of unknown data point from all the points in the dataset

Hence, we have calculated the Euclidean distance of unknown data point from all the points as shown:

Where (x1, y1) = (57, 170) whose class we have to classify

Weight(x2)	Height(y2)	Class	Euclidean Distance
51	167	Underweight	6.7
62	182	Normal	13
69	176	Normal	13.4
64	173	Normal	7.6
65	172	Normal	8.2
56	174	Underweight	4.1
58	169	Normal	1.4
57	173	Normal	3
55	170	Normal	2

Now, lets calculate the nearest neighbor at k=3

Weight(x2)	Height(y2)	Class	Euclidean Distance
51	167	Underweight	6.7
62	182	Normal	13
69	176	Normal	13.4
64	173	Normal	7.6
65	172	Normal	8.2
56	174	Underweight	4.1
58	169	Normal	1.4
57	173	Normal	3
55	170	Normal	2

k = 3

57 kg	**170 cm**	**?**

Class	Euclidean Distance
Underweight	6.7
Normal	13
Normal	13.4
Normal	7.6
Normal	8.2
Underweight	4.1
Normal	1.4
Normal	3
Normal	2

k = 3

So, majority neighbors are pointing towards '*Normal*'

Hence, as per KNN algorithm the class of (57, 170) should be 'Normal'

Pros of kNN

- It is very simple algorithm to understand and interpret.
- It is very useful for nonlinear data because there is no assumption about data in this algorithm.
- It is a versatile algorithm as we can use it for classification as well as regression.
- It has relatively high accuracy but there are much better supervised learning models than KNN.

Cons of kNN

- It is computationally a bit expensive algorithm because it stores all the training data.
- High memory storage required as compared to other supervised learning algorithms.
- Prediction is slow in case of big N.
- It is very sensitive to the scale of data as well as irrelevant features.

Decision Tree

- Decision Tree is a **Supervised learning technique** that can be used for both classification and Regression problems, but mostly it is preferred for solving Classification problems.

- It is a tree-structured classifier, where **internal nodes represent the features of a dataset, branches represent the decision rules** and **each leaf node represents the outcome.**

- In a Decision tree, there are two nodes, which are the **Decision Node** and **Leaf Node.**

- Decision nodes are used to make any decision and have multiple branches, whereas **Leaf nodes are the output of those decisions and do not contain any further branches.**

- The decisions or the **test are performed on the basis of features of the given dataset.**

- ***It is a graphical representation for getting all the possible solutions to a problem/decision based on given conditions.***

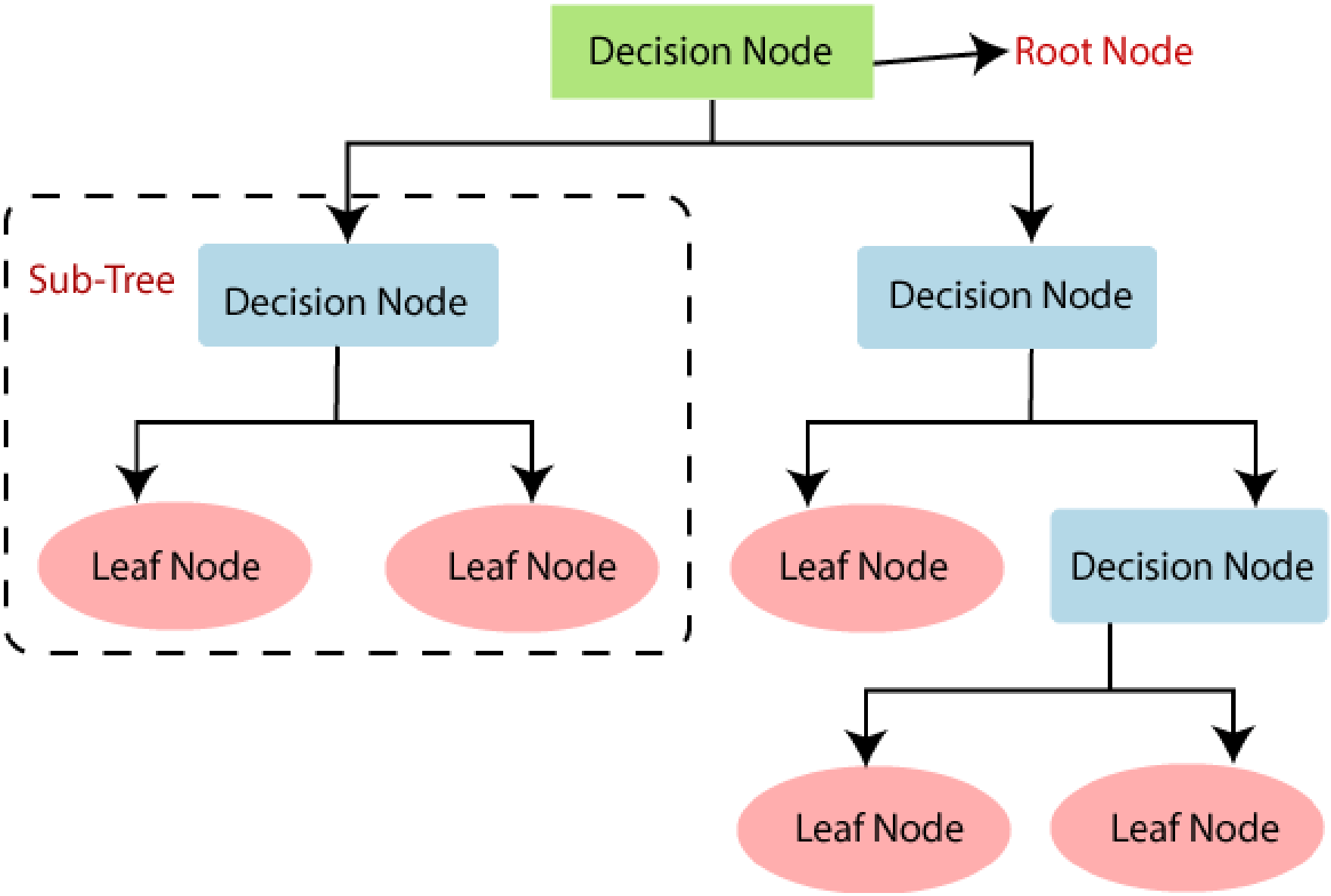
- It is called a decision tree because, similar to a tree**, it starts with the root node**, which **expands on further branches** and **constructs a tree-like structure.**

- In order to build a tree, we use the **CART algorithm,** which stands for **Classification and Regression Tree algorithm.**

- A decision tree simply asks a question, and **based on the answer (Yes/No), it further split the tree into subtrees.**

- **A decision tree can contain categorical data (YES/NO) as well as numeric data.**

General Structure of a Decision Tree



Why use Decision Trees?

- There are various algorithms in Machine learning, so choosing the best algorithm for the given dataset and problem is the main point to remember while creating a machine learning model.

- Below are the two reasons for using the Decision tree:
  - Decision Trees **usually copy human thinking ability while making a decision, so it is easy to understand.**

  - The logic behind the decision tree **can be easily understood because it shows a tree-like structure.**

Decision Tree Terminologies

- **Root Node:** Root node is from where the **decision tree starts**. It represents the entire dataset, which further gets divided into two or more homogeneous sets.

- **Leaf Node:** Leaf nodes are the **final output node**, and the tree cannot be segregated further after getting a leaf node.

- **Splitting:** Splitting is the process of **dividing the decision node/root node** into sub-nodes according to the given conditions.

- **Branch/Sub Tree:** A tree formed by **splitting the tree**.

- **Pruning:** Pruning is the process **of removing the unwanted branches from the tree.**

- **Parent/Child node:** The **root node of the tree is called the parent node**, and other nodes are called the child nodes.

How does the Decision Tree algorithm Work?

- In a decision tree, for predicting the class of the given dataset, **the algorithm starts from the root node of the tree.**

- This algorithm **compares the values of root attribute with the record (real dataset) attribute** and, based on the comparison, follows the branch and jumps to the next node.

- For the next node, the **algorithm again compares the attribute value with the other sub-nodes** and move further. **It continues the process until it reaches the leaf node of the tree.**

Steps for the Decision Tree

- **Step-1:** Begin the tree with the root node, says S, which contains the complete dataset.
- **Step-2:** Find the best attribute in the dataset using **Attribute Selection Measure (ASM).**
- **Step-3:** Divide the S into subsets that contains possible values for the best attributes.
- **Step-4:** Generate the decision tree node, which contains the best attribute.
- **Step-5:** Recursively make new decision trees using the subsets of the dataset created in step -3.
- Continue **this process until a stage is reached where you cannot further classify the nodes and called the final node as a leaf node.**

Attribute Selection Measures

- While implementing a Decision tree, the **main issue arises that how to select the best attribute for the root node and for sub-nodes.**

- So, to solve such problems there is a technique which is called as **Attribute selection measure or ASM.**

- By this measurement, **we can easily select the best attribute for the nodes of the tree.**

- There are two popular techniques for ASM, which are:
  - **Information Gain**
  - **Gini Index**

1. Information Gain

- Information gain is a decrease in entropy. It **computes the difference between entropy before split and average entropy after split of the dataset based on given attribute values.**
- It **calculates how much information a feature provides us about a class.**
- According to the value of information gain, **we split the node and build the decision tree.**
- A decision tree algorithm always tries to **maximize the value of information gain**, and a node/attribute having the highest information gain is split first.
- It can be calculated using the below formula:
- **Information Gain= Entropy(S)- [(Weighted Avg) * Entropy(each feature)**

- **If the information gain 0**, it means that **there is no reduction in entropy** due to split of the data set according to that particular feature.

- **Maximum amount of information gain which may happen is the entropy of the data set before the split.**

Entropy

- Entropy is a metric to **measure the impurity in a given attribute**.
- It specifies randomness in data.

- Entropy can be calculated as:
- **Entropy(s)= -P(yes)log2 P(yes)- P(no) log2 P(no)**

- **Where,**
  - **S=Total number of samples**
  - **P(yes)= probability of yes**
  - **P(no)= probability of no**

2. Gini Index

- Gini index is a **measure of impurity or purity used while creating a decision tree in the CART(Classification and Regression Tree) algorithm.**

- An attribute with the **low Gini index should be preferred as compared to the high Gini index.**

- It only **creates binary splits**, and the **CART algorithm uses the Gini index to create binary splits.**

- Gini index can be calculated using the below formula:
- **Gini Index= 1-** $\sum_j P_j^2$

Pruning: Getting an Optimal Decision tree

- *Pruning is a* ***process of deleting the unnecessary nodes from a tree in order to get the optimal decision tree.***

- A **too-large tree increases the risk of overfitting**, and **a small tree may not capture all the important features of the dataset.**

- Therefore, a technique that decreases the size of the learning tree without reducing accuracy is known as Pruning.

- There are mainly two types of tree **pruning** technology used:

  - **Cost Complexity Pruning**
  - **Reduced Error Pruning.**

Advantages of the Decision Tree

- It is simple to understand as it follows the same process which a human follow while making any decision in real-life.
- It can be very useful for solving decision-related problems.
- It helps to think about all the possible outcomes for a problem.
- There is less requirement of data cleaning compared to other algorithms.

Disadvantages of the Decision Tree

- The decision tree contains lots of layers, which makes **it complex.**
- It may have **an overfitting issue**, which can be resolved using the **Random Forest algorithm.**
- For **more class labels**, **the computational complexity** of the decision tree may increase.

Random Forest Algorithm

- Random Forest is a popular machine learning algorithm **that belongs to the supervised learning technique.**

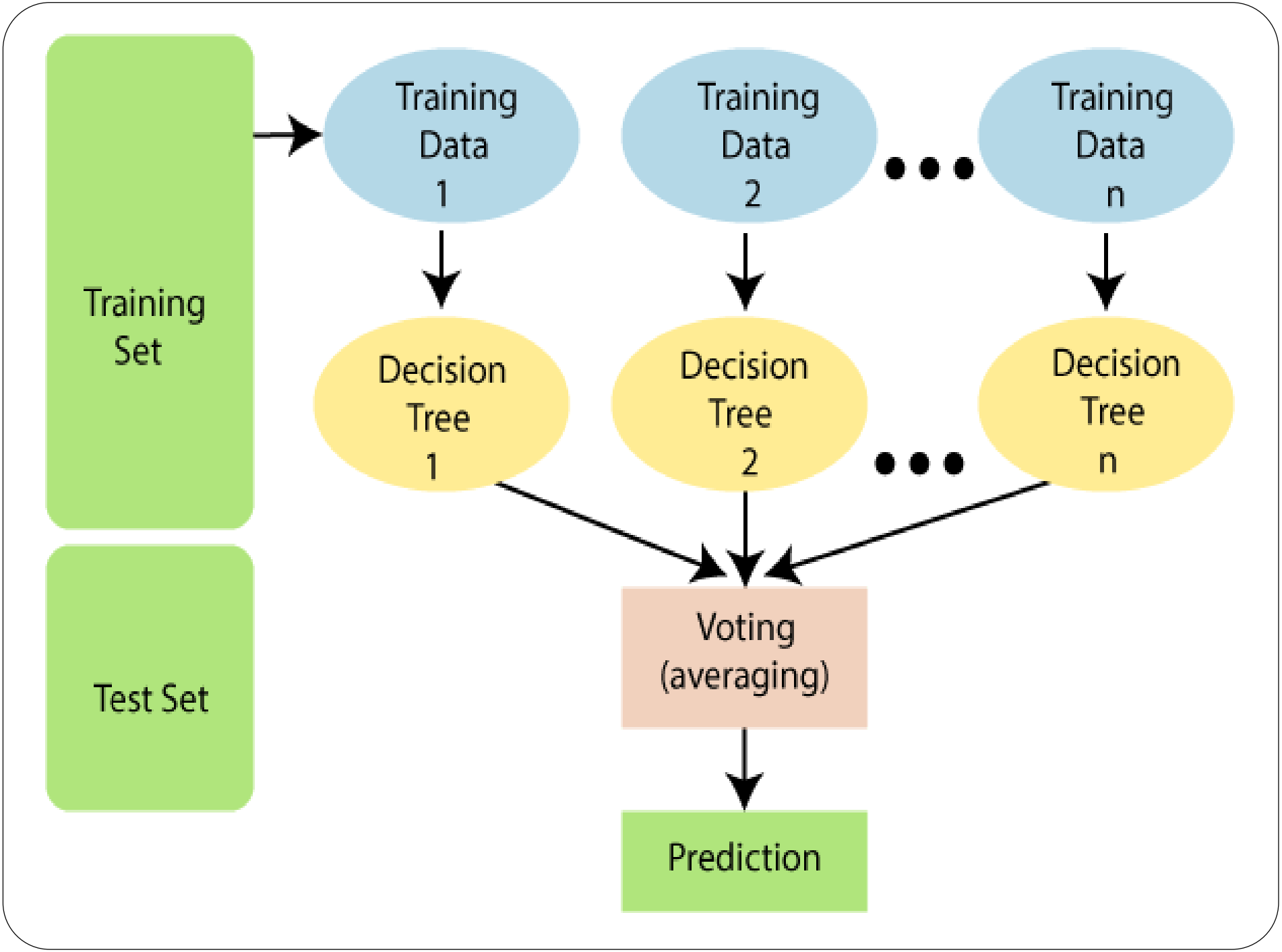
- It can be **used for both Classification and Regression** problems in ML.

- It is based on the concept of **ensemble (collection) learning,** which is a process of ***combining multiple classifiers to solve a complex problem*** *and* ***to improve the performance of the model***.

- Random Forest is a classifier that **contains a number of decision trees on various subsets of the given dataset and takes the average to improve the predictive accuracy of that dataset.**

- Instead of relying on one decision tree**, the random forest takes the prediction from each tree and based on the majority votes of predictions, and it predicts the final output.**

- The greater number of trees in the forest leads to **higher accuracy and prevents the problem of overfitting.**


Training Set
Test Set
Training Data 1
Training Data 2
Training Data n
Decision Tree 1
Decision Tree 2
Decision Tree n
Voting (averaging)
Prediction

Why use Random Forest?

- It **takes less training time** as compared to other algorithms.
- It predicts output **with high accuracy**, even for the large dataset it runs efficiently.
- It can also maintain accuracy when a large proportion of data is missing.

How does Random Forest algorithm work?

- Random Forest works in **two-phase first is to create the random forest by combining N decision tree**, and **second is to make predictions for each tree created in the first phase.**

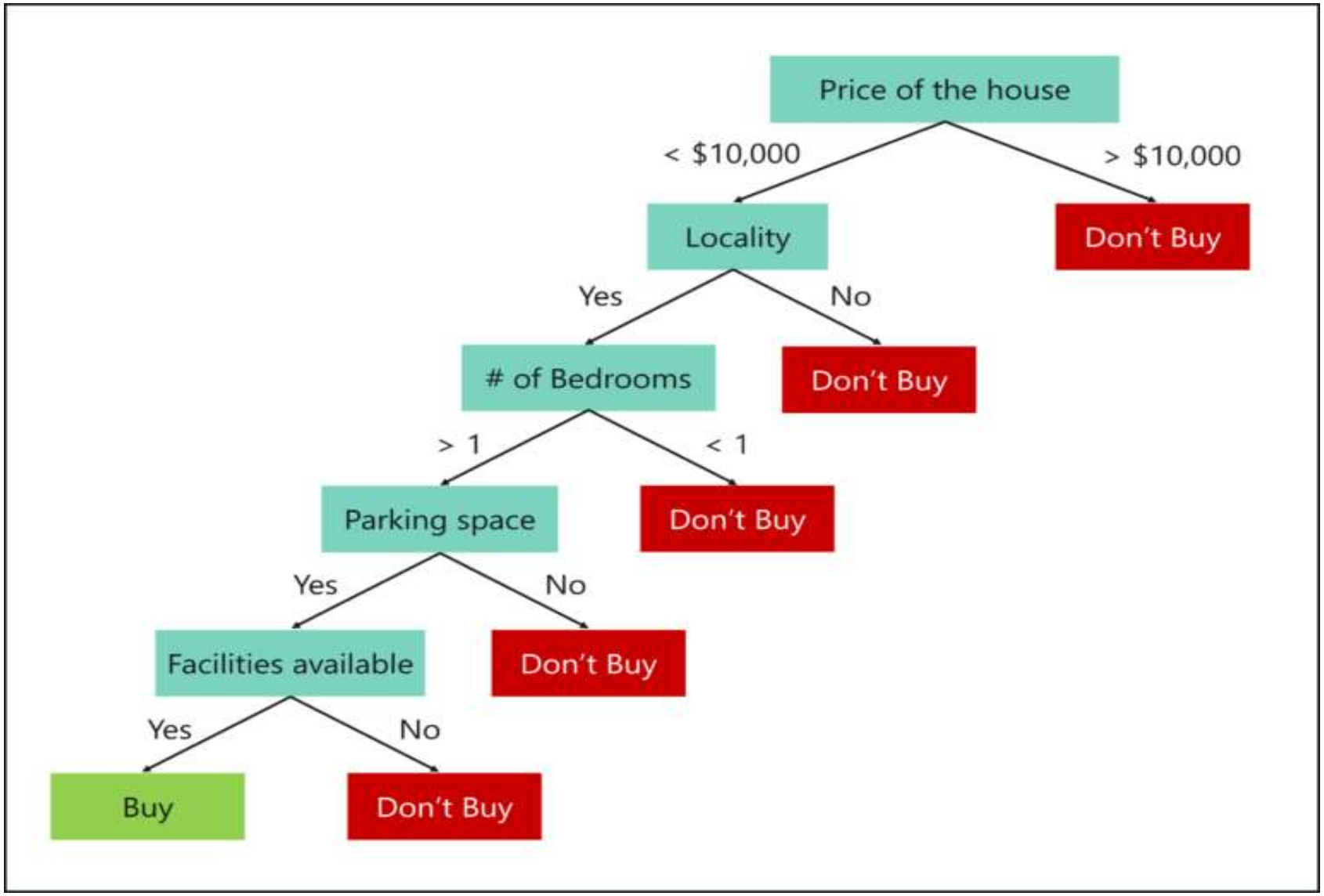
- The Working process can be explained in the below steps and diagram:

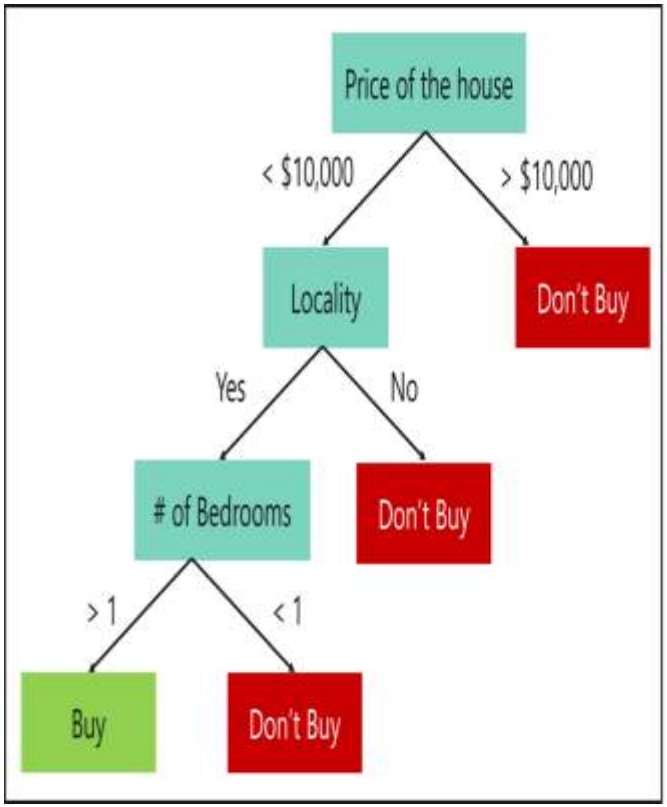
  - **Step-1:** Select random K data points from the training set.
  - **Step-2:** Build the decision trees associated with the selected data points (Subsets).
  - **Step-3:** Choose the number N for decision trees that you want to build.
  - **Step-4:** Repeat Step 1 & 2.
  - **Step-5:** For new data points, find the predictions of each decision tree, and assign the new data points to the category that wins the majority votes.

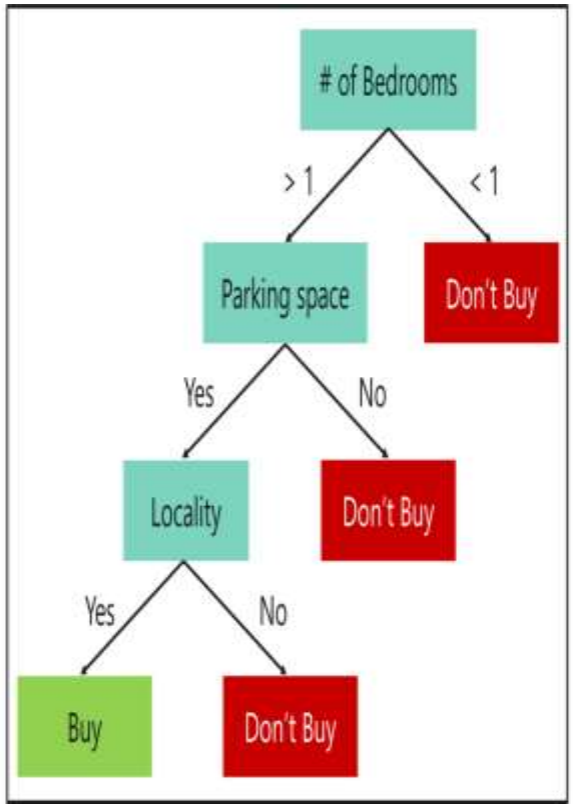
Example of Random Forest

- you're looking to buy a house, but you're unable to decide which one to buy. So, you consult a few agents and they give you a list of parameters that you should consider before buying a house.
- The list includes:
  - **Price of the house**
  - **Locality**
  - **Number of bedrooms**
  - **Parking space**
  - **Available facilities**
- These parameters are known as *predictor variables*, which are used to find the response variable.

Decision Tree for example




Price of the house
< $10,000
> $10,000
Locality
Don't Buy
Yes
No
# of Bedrooms
Don't Buy
> 1
< 1
Buy
Don't Buy


# of Bedrooms
> 1
< 1
Parking space
Don't Buy
Yes
No
Locality
Don't Buy
Yes
No
Buy
Don't Buy

Locality
Yes
No
Price of the house
Don't Buy
< $10,000
> $10,000
Facilities available
Don't Buy
Yes
No
Buy
Don't Buy

- **Decision trees** are built on the **entire data set** using all the predictor variables, **whereas Random Forests** are used to create **multiple decision trees**, such that **each decision tree is built only on a part of the data set.**

Applications of Random Forest

- **Banking:** Banking sector mostly uses this algorithm for the identification of loan risk.

- **Medicine:** With the help of this algorithm, disease trends and risks of the disease can be identified.

- **Land Use:** We can identify the areas of similar land use by this algorithm.

- **Marketing:** Marketing trends can be identified using this algorithm.

- ## Advantages of Random Forest

  - Random Forest is capable of **performing both Classification and Regression tasks.**

  - It is capable of **handling large datasets** with high dimensionality.

  - It **improves the accuracy** of the model and prevents the overfitting issue.

- ## Disadvantages of Random Forest

  - Although random forest can be used for both classification and regression tasks, **it is not more suitable for Regression tasks.**

Support Vector Machine Algorithm

- Support Vector Machine or SVM is one of the **most popular Supervised Learning algorithms, which is used for Classification as well as Regression problems.**
- However, **primarily, it is used for Classification problems in Machine Learning.**
- The goal of the SVM algorithm is to create the **best line or decision boundary that can segregate n-dimensional space into classes so that we can easily put the new data point in the correct category in the future.**
- This **best decision boundary is called a hyperplane.**
- **SVM chooses the extreme points/vectors that help in creating the hyperplane.**
- These **extreme cases are called as support vectors, and hence algorithm is termed as Support Vector Machine.**

Types of SVM

- **SVM can be of two types:**
- **Linear SVM:** Linear SVM is used for linearly separable data, which means if **a dataset can be classified into two classes by using a single straight line**, then such data is termed as linearly separable data, and **classifier is used called as Linear SVM classifier**.

- **Non-linear SVM:** Non-Linear SVM is used for non-linearly separated data, which means if a **dataset cannot be classified by using a straight line**, then such data is termed as non-linear data and classifier used is called as Non-linear SVM classifier.

Important Terminology

- **Hyperplane:** There can be multiple lines/decision boundaries to segregate the classes in n-dimensional space, but **we need to find out the best decision boundary that helps to classify the data points**. This best boundary is known as the hyperplane of SVM.

- The dimensions of the hyperplane depend on the features present in the dataset, which means if there are 2 features (as shown in image), then hyperplane will be a straight line. And if there are 3 features, then hyperplane will be a 2-dimension plane.
- We always create a hyperplane that has a maximum margin, which means the maximum distance between the data points.

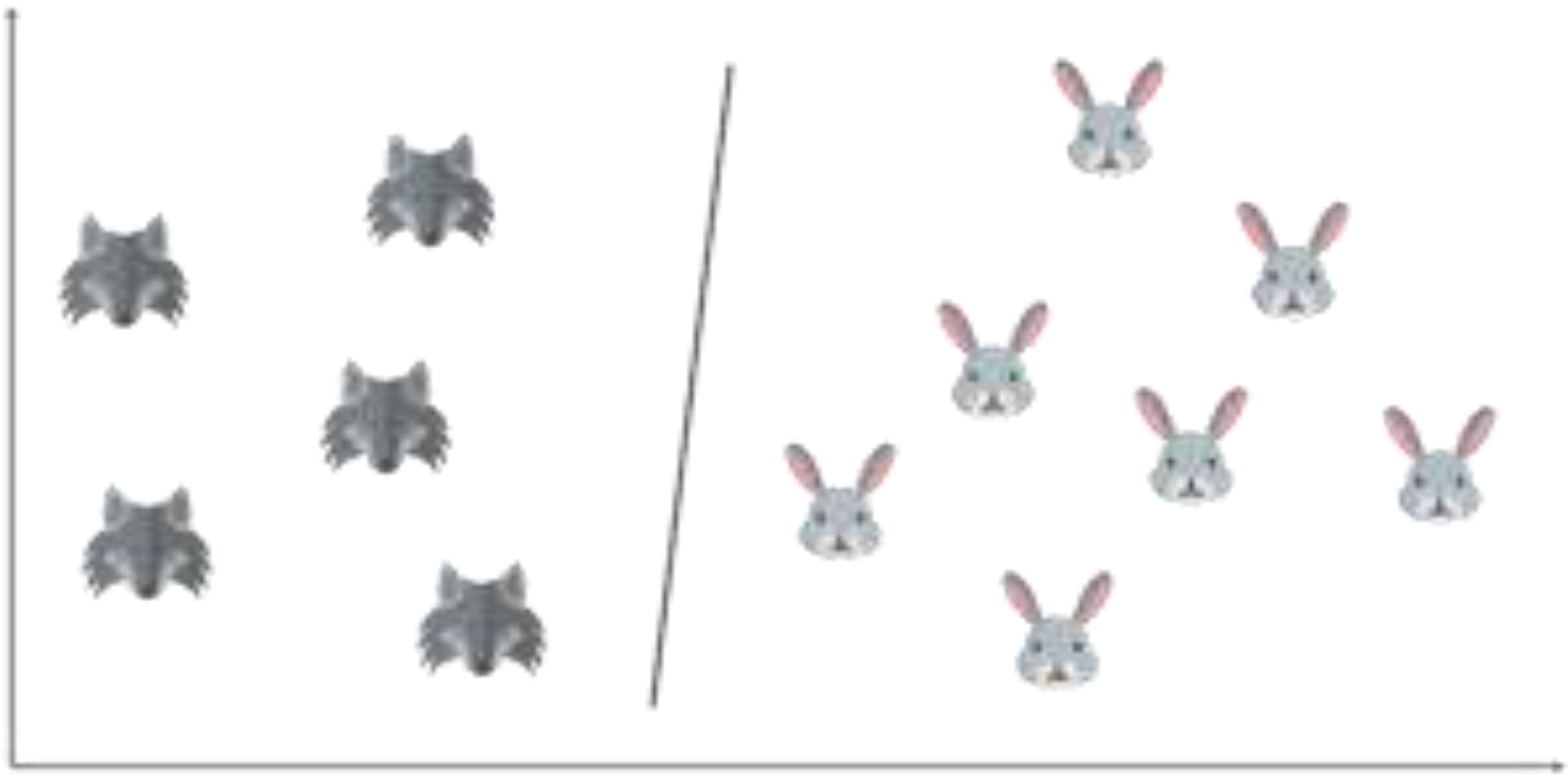
- **Support Vectors:**
- The data points or vectors that are the closest to the hyperplane and which affect the position of the hyperplane are termed as Support Vector. Since these vectors support the hyperplane, hence called a Support vector.

- **Kernel trick:** SVM can be used for **classifying non-linear data by using the kernel trick**.
- The kernel trick **means transforming data into another dimension that has a clear dividing margin between classes of data**.
- After which you can easily draw a hyperplane between the various classes of data.
- It **convert linearly non-seprable data to a linearly separable data that's why its called karnels.**

How Does SVM Work?

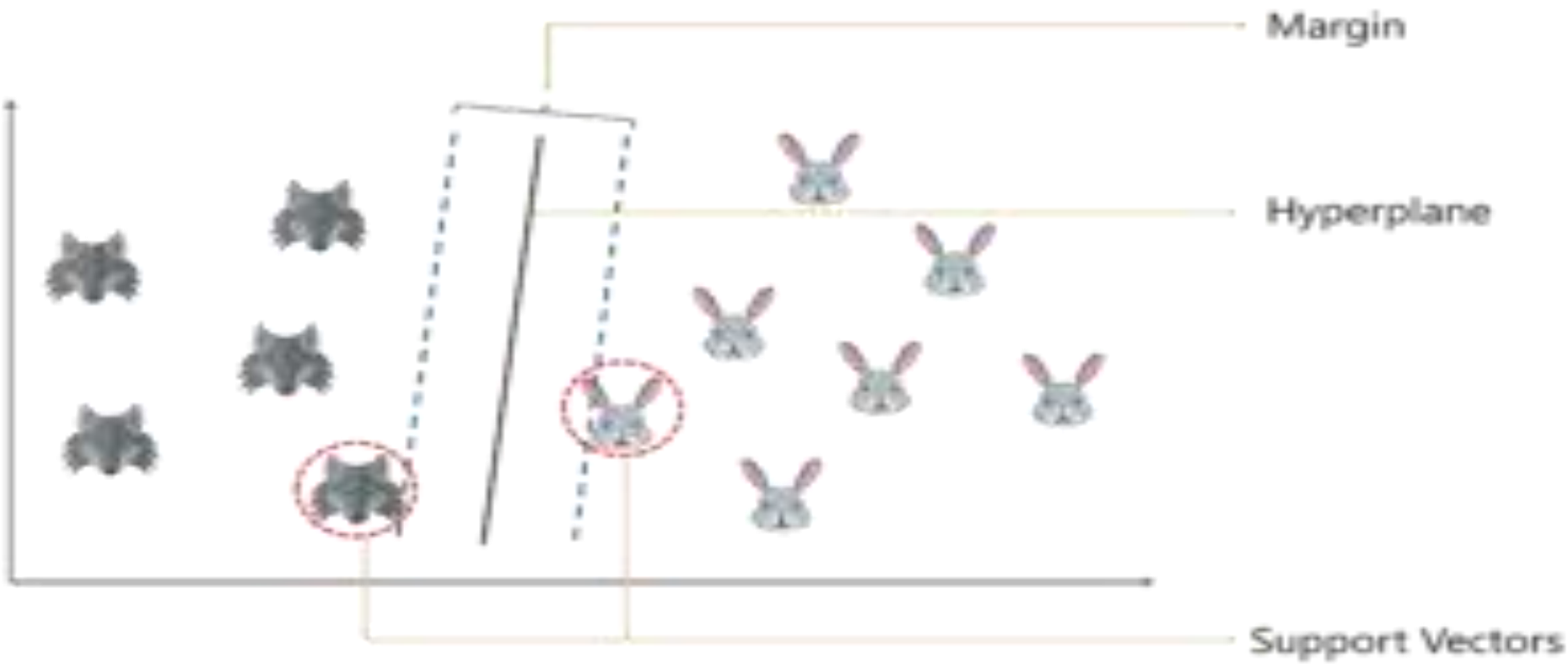
- Imagine you own a farm and you have a problem–you need to set up a fence to protect your rabbits from a pack of wolves. But where do you build your fence?

- draw a decision boundary between the rabbits and the wolves, it looks something like this. Now you can clearly build a fence along this line.
- This is exactly how SVM works. It **draws a decision boundary**, i.e. a **hyperplane between any two classes in order to separate them or classify them.**



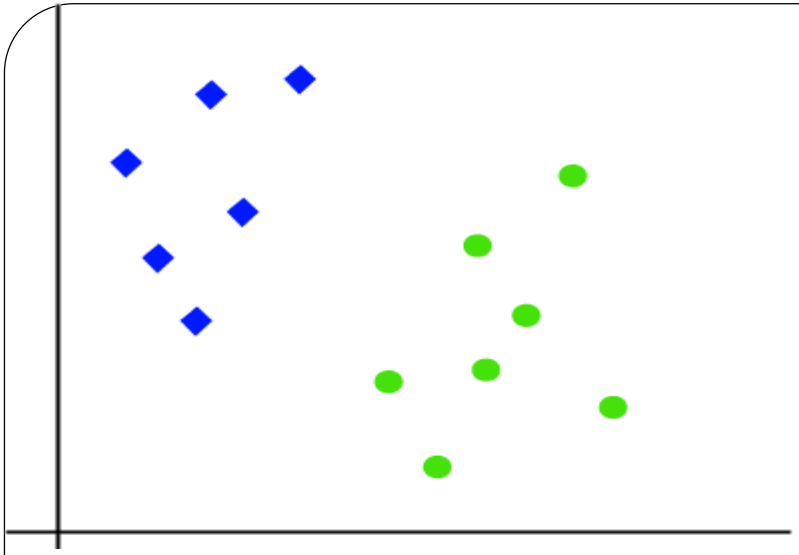
What is a Support Vector in SVM?

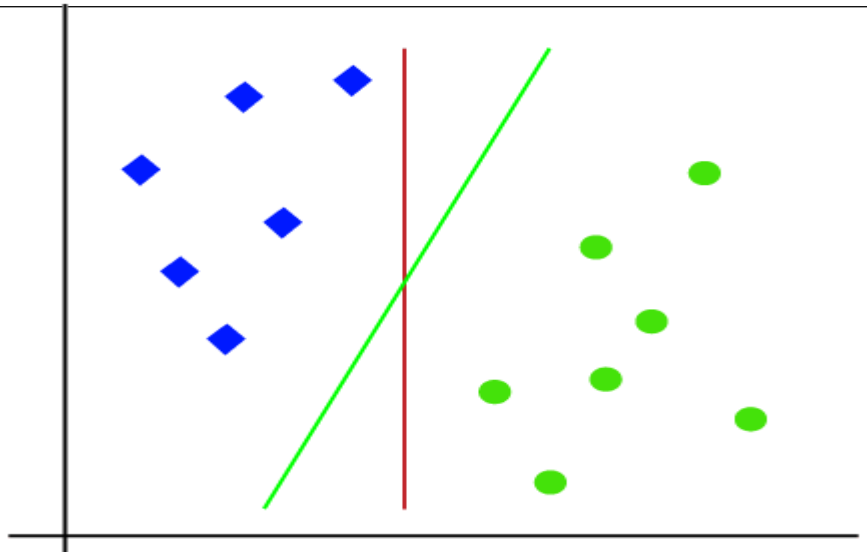
- drawing a random hyperplane and then you **check the distance between the hyperplane and the closest data points from each class**.
- These **closest data points to the hyperplane are known as support vectors.**

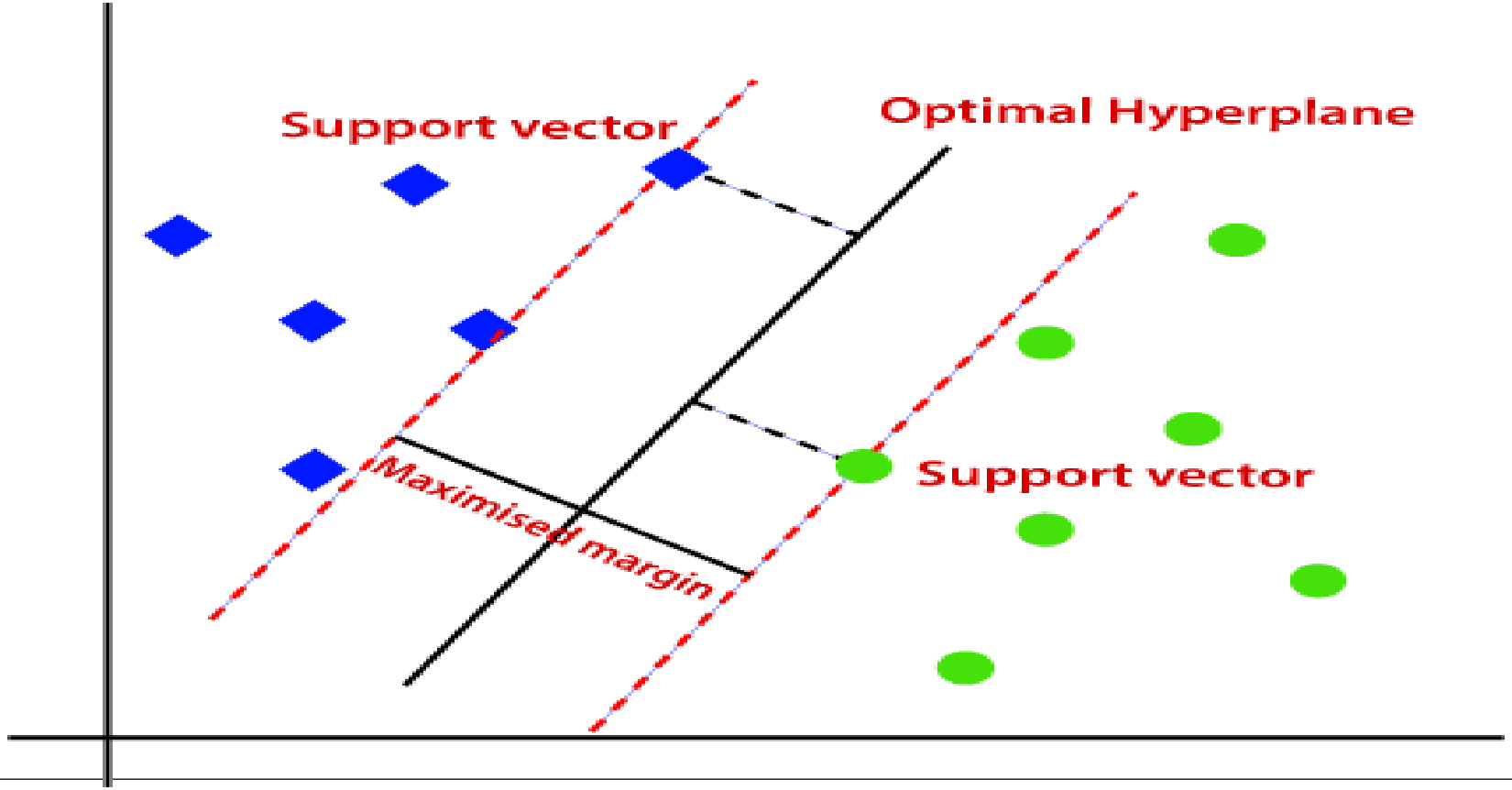


- The hyperplane is drawn based on these support vectors and an optimum hyperplane will have a maximum distance from each of the support vectors. And this **distance between the hyperplane and the support vectors is known as the margin.**

- **To sum it up**, SVM is used to **classify data by using a hyperplane, such that the distance between the hyperplane and the support vectors is maximum.**



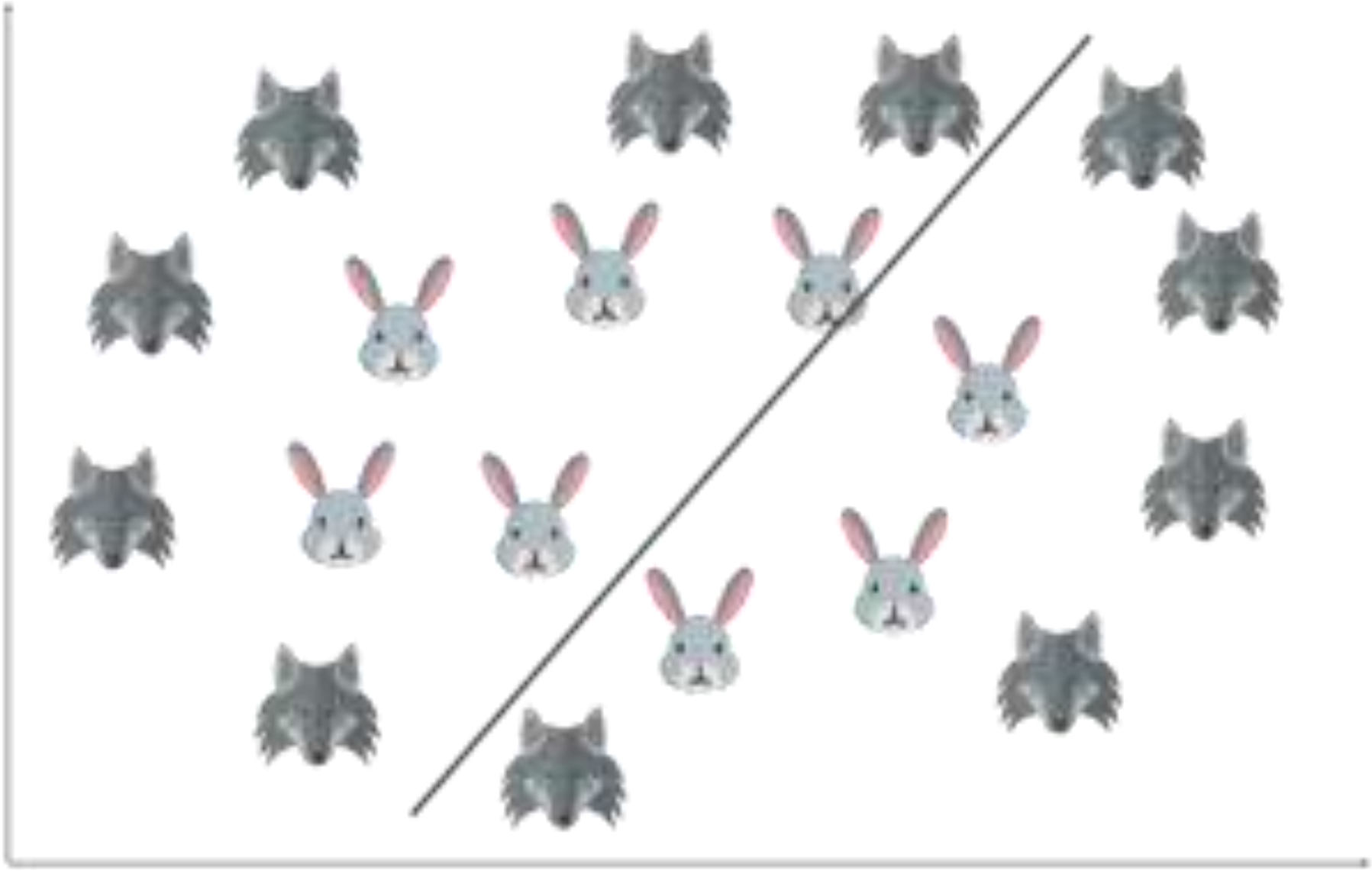



Support vector
Optimal Hyperplane
Support vector
Maximised margin

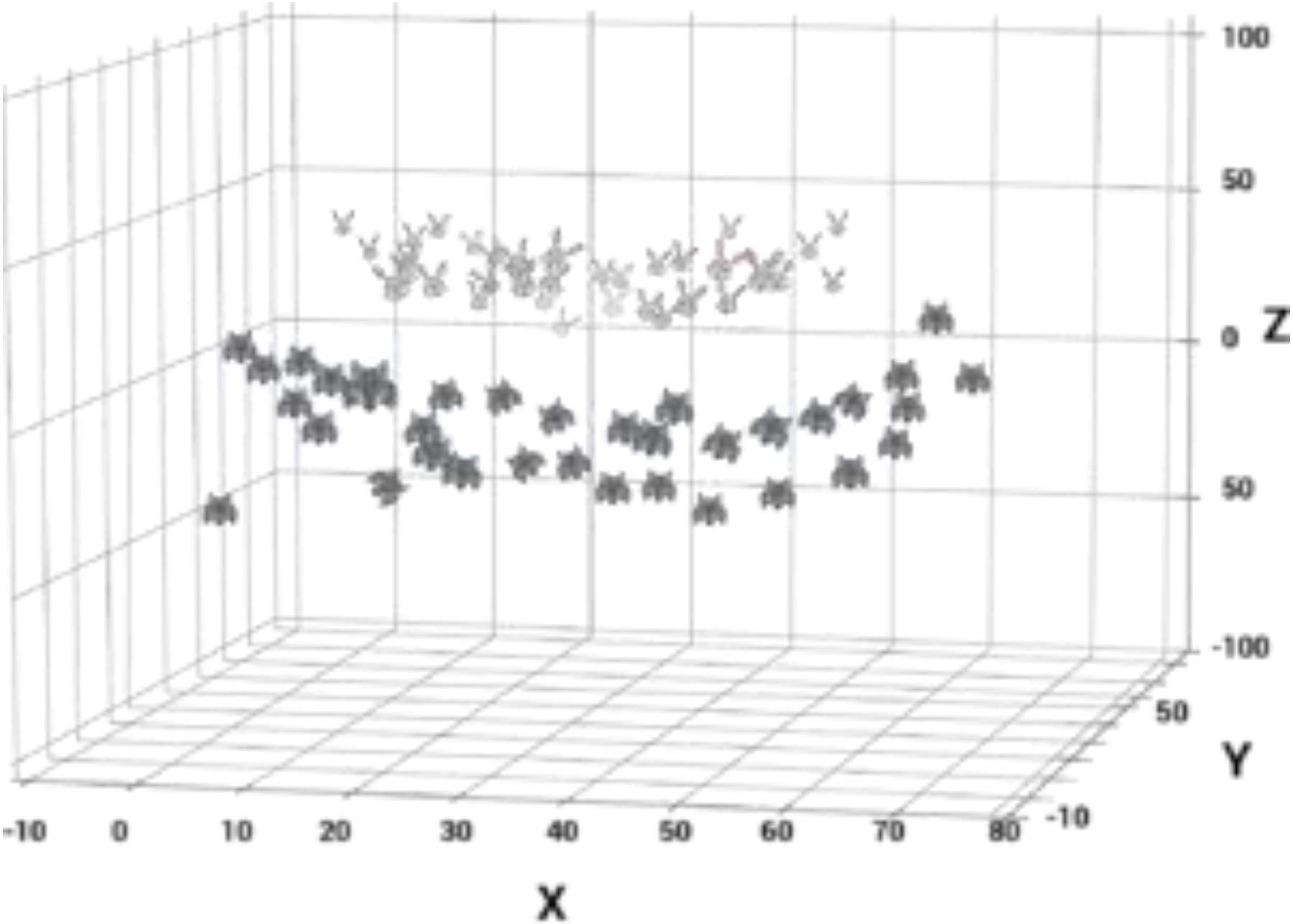
Non-Linear SVM

- **what will you do if the data set is like this?**

- You possibly can't draw a hyperplane like this! It doesn't separate the two classes.



- **kernel** can be used to **transform data into another dimension that has a clear dividing margin between classes of data.**
- Kernel functions offer the user the option of **transforming nonlinear spaces into linear ones.**
- Until this point, we were plotting our data on 2-dimensional space. So, we had only 2 variables, x and y.
- A simple trick would be transforming the two variables x and y into a new feature space involving a **new variable z**. Basically, we're visualizing the data on a 3-dimensional space.
- **When you transform the 2D space into a 3D space you can clearly see a dividing margin between the 2 classes of data**. And now you can go ahead and **separate the two classes by drawing the best hyperplane between them.**


100
50
0
Z
50
-100
50
Y
-10
-10
0
10
20
30
40
50
60
70
80
X

Advantages of SVM

- **Effective in high dimensional spaces.**
- Still effective in cases where the number of dimensions is greater than the number of samples.
- Uses a subset of training points in the decision function **that makes it memory efficient.**
- **Different kernel functions** can be specified for the decision function that also makes it versatile.

Disadvantages of SVM

- If the number of features is much larger than the number of samples, avoid over-fitting in **choosing kernel functions and regularization term is crucial.**

- **SVMs do not directly provide probability estimates**, these are calculated using five-fold cross-validation.

Support Vector Machine Use Cases

- **Face Detection**
- **Text And HyperText Categorization**
- **Classification Of Images**
- **Bioinformatics**
- **Protein Fold and Remote Homology Detection**
- **Handwriting Recognition**
- **Generalized Predictive Control**

Example of regression

- **Common Regression Algorithms**
  - **Simple linear regression**
  - Multiple linear regression
  - Polynomial Regression
  - Multivariate adaptive regression splines
  - **Logistic regression**
  - Maximum likelihood estimation (least squares)

Simple linear Regression

- **What is Regression?**
- The main goal of regression is the **construction of an efficient model to predict the dependent attributes from a bunch of attribute variables.**
- A regression problem is **when the output variable is either real or a continuous value i.e salary, weight, area, etc.**
- We can also define regression as a statistical means that is used in applications like housing, investing, etc.
- It is **used to predict the relationship between a dependent variable and a bunch of independent variables**.

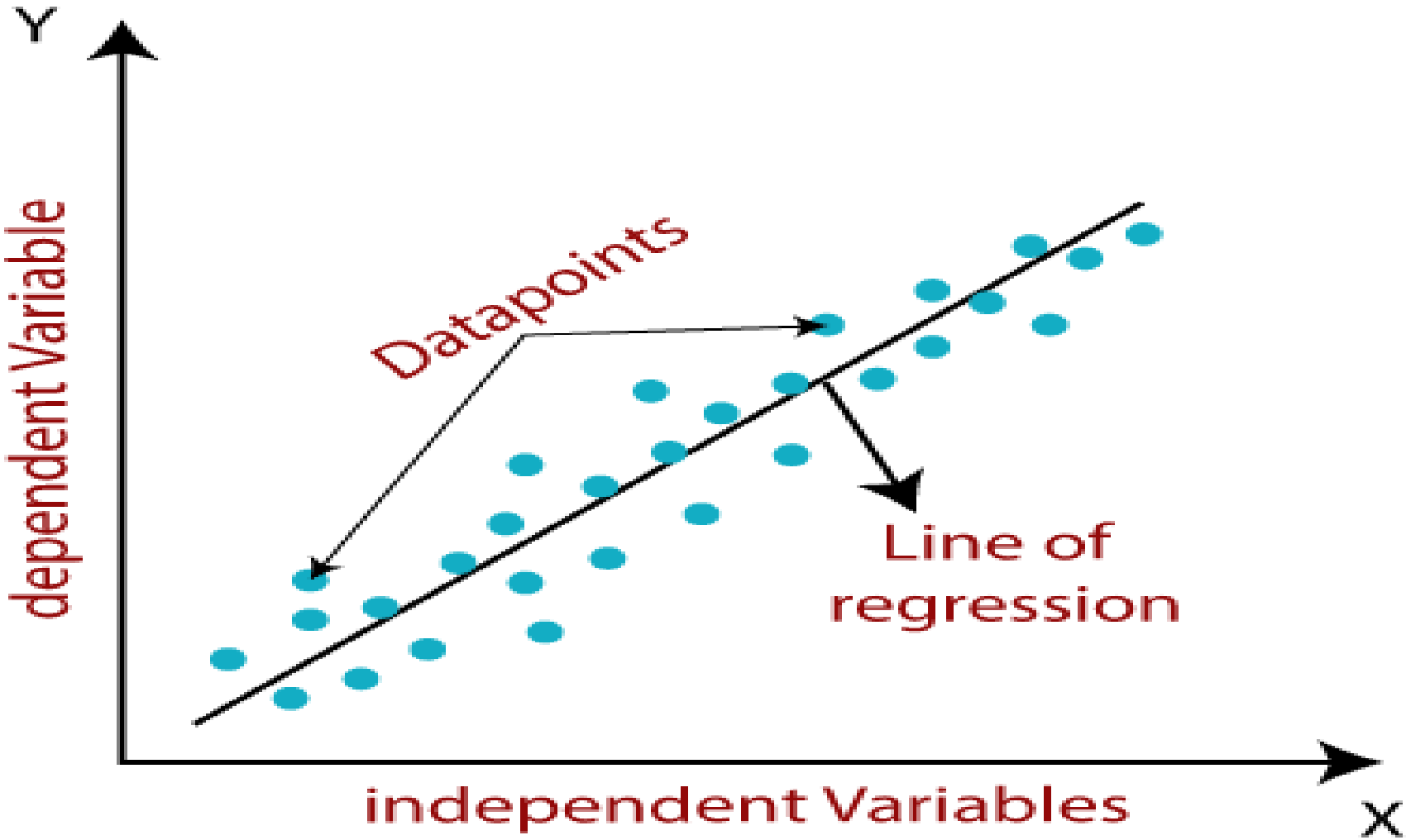
- **Dependent Variable** – A Dependent Variable is the variable to be predicted or explained in a regression model. **This variable is assumed to be functionally related to the independent variable**.

- **Independent Variable** – An Independent Variable is the variable related to the dependent variable in a regression equation. The independent variable is used in a regression model to **estimate the value of the dependent variable.**

Linear Regression

- Linear regression is one of the easiest and most popular Machine Learning algorithms.
- It is a statistical method that is **used for predictive analysis**.
- Linear regression makes predictions for continuous/real or numeric variables such as **sales, salary, age, product price,** etc.
- Linear regression algorithm shows a **linear relationship between a dependent (y) and one or more independent (y) variables, hence called as linear regression.**
- Since linear regression shows the linear relationship, which **means it finds how the value of the dependent variable is changing according to the value of the independent variable.**

- The linear regression model provides a **sloped straight line representing the relationship between the variables.**



- Mathematically, we can represent a linear regression as:
  - $\mathbf{y = a_0 + a_1 x + \varepsilon}$
- **Here,**
- Y= Dependent Variable (Target Variable)
  X= Independent Variable (predictor Variable)
  a0= intercept of the line (Gives an additional degree of freedom)
  a1 = Linear regression coefficient (scale factor to each input value).
  ε = random error

- The values for x and y variables are training datasets for Linear Regression model representation.

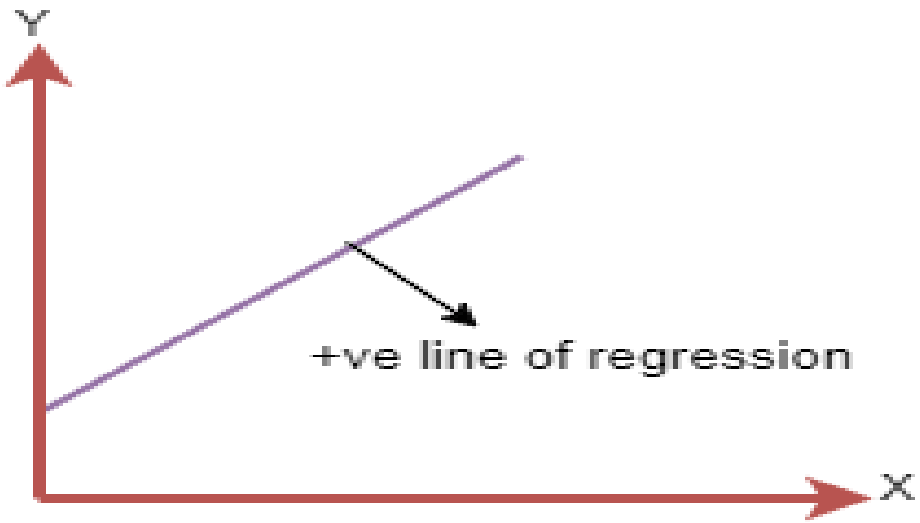
Types of Linear Regression

- **Simple Linear Regression:**
  If a **single independent variable** is used to **predict the value of a numerical dependent variable, then such a Linear Regression algorithm is called Simple Linear Regression.**

- **Multiple Linear regression:**
  If **more than one independent variable** is used to **predict the value of a numerical dependent variable, then such a Linear Regression algorithm is called Multiple Linear Regression**.

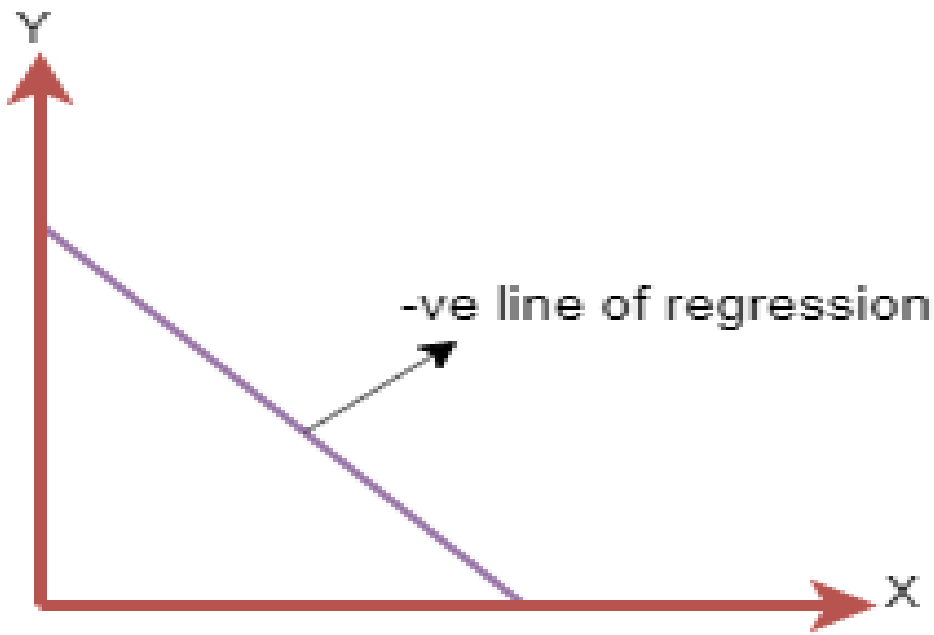
Linear Regression Line

- A linear line showing the relationship between the dependent and independent variables is called a **regression line**.
- A regression line can show two types of relationship:
- **Positive Linear Relationship:**
  - If the **dependent variable increases on the Y-axis** and **independent variable increases on X-axis**, then such a relationship is termed as a **Positive linear relationship**.



The line equation will be: $Y = a_0 + a_1 x$

- **Negative Linear Relationship:**
- If the **dependent variable decreases on the Y-axis** and **independent variable increases on the X-axis**, then such a relationship is called a negative linear relationship.



The line of equation will be: $Y = -a_0 + a_1 x$

Finding the best fit line

- When working with linear regression, our **main goal is to find the best fit line** that means **the error between predicted values and actual values should be minimized.**

- **The best fit line will have the least error.**

- The different values for weights or the coefficient of lines (a_0, a_1) gives a different line of regression, **so we need to calculate the best values for a_0 and a_1 to find the best fit line, so to calculate this we use cost function.**

- The **cost function** **provides the best possible values for b_0 and b_1 to make the best fit line for the data points.**
- We do it by converting this problem into a **minimization problem to get the best values for b_0 and b_1.**
- The error is minimized in this problem between the actual value and the predicted value.

$$minimize \frac{1}{n} \sum_{i=1}^{n} (pred_i - y_i)^2$$

$$J = \frac{1}{n} \sum_{i=1}^{n} (pred_i - y_i)^2$$

- We choose the **function above to minimize the error.**
- We square the error difference and **sum the error over all data points, the division between the total number of data points.**
- Then, the produced value provides the averaged square error over all data points.
- It is also known as **MSE(Mean Squared Error)**, and **we change the values of b_0 and b_1 so that the MSE value is settled at the minimum**.

Gradient Descent

- The next important terminology to understand linear regression is **gradient descent.** It is a method of updating b_0 and b_1 values to reduce the MSE.

- The idea behind this is to keep iterating the b_0 and b_1 values **until we reduce the MSE to the minimum.**

Model Performance

- The Goodness of fit determines how the line of regression fits the set of observations.
- The process of finding the best model out of various models is called **optimization**. It can be achieved by below method:
- **1. R-squared method:**
  - R-squared is a statistical method that determines the goodness of fit.
  - It measures the strength of the relationship between the dependent and independent variables on a scale of 0-100%.
  - The high value of R-square determines the less difference between the predicted values and actual values and hence represents a good model.
  - It is also called a **coefficient of determination,** or **coefficient of multiple determination** for multiple regression.

- It can be calculated from the below formula:

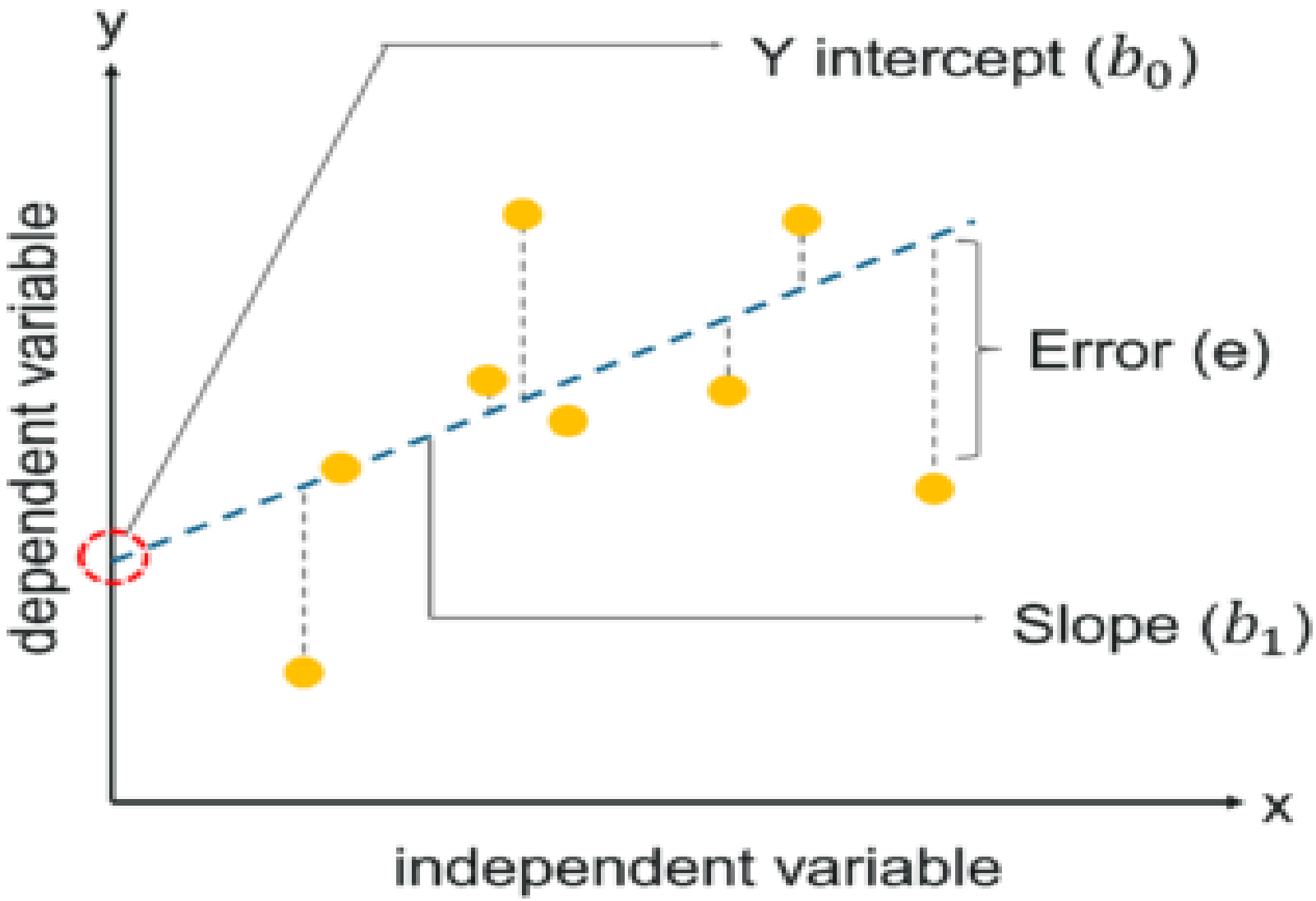
$$\text{R-squared} = \frac{\text{Explained variation}}{\text{Total Variation}}$$

Simple Linear Regression

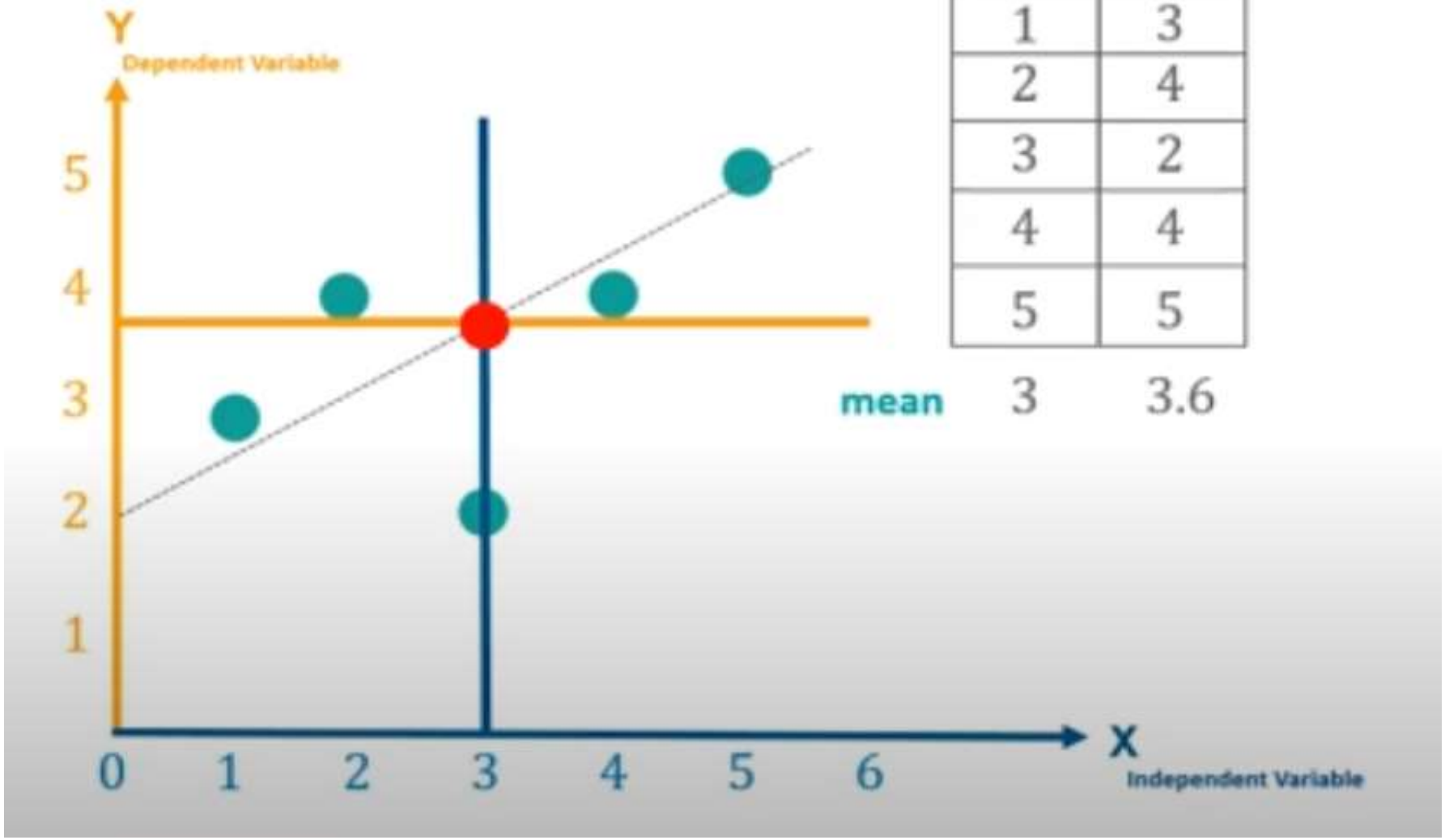
- Simple Linear Regression is a type of Regression algorithms that models the **relationship between a dependent variable and a single independent variable.**
- The relationship shown by a Simple Linear Regression model is linear or a sloped straight line, hence it is called Simple Linear Regression.
- The key point in Simple Linear Regression is that the ***dependent variable must be a continuous/real value***.
- However, the **independent variable can be measured on continuous or categorical values.**
- Simple Linear regression algorithm has mainly two objectives:
  - **Model the relationship between the two variables.** Such as the relationship between Income and expenditure, experience and Salary, etc.
  - **Forecasting new observations.** Such as Weather forecasting according to temperature, Revenue of a company according to the investments in a year, etc.

- Simple Linear Regression Model:
- The Simple Linear Regression model can be represented using the below equation:
  - $\mathbf{y = a_0 + a_1 x + \varepsilon}$
- Where,
- **a0=** It is the intercept of the Regression line (can be obtained putting x=0)
- **a1=** It is the slope of the regression line, which tells whether the line is increasing or decreasing.
- **ε =** The error term. (For a good model it will be small)

Linear Regression



Example of simple regression

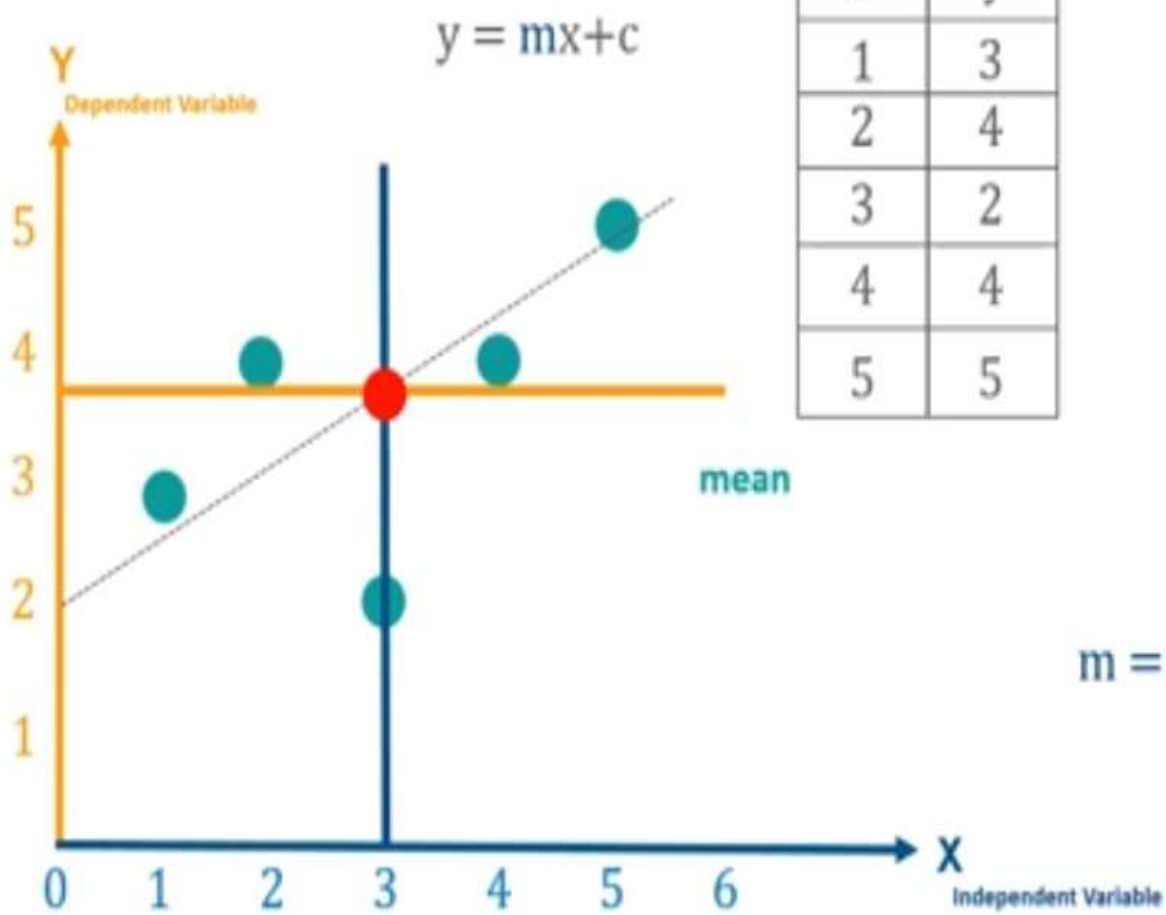


x	y
1	3
2	4
3	2
4	4
5	5
mean 3	3.6

Find equation of line

- **Equation for a Straight Line**
- We begin with the equation $Y = a + b X$
  - Y = Dependent variable
  - A = Y-intercept
  - B = Slope of the line
  - X = Independent variable
- Y & X are linearly independent and linearly correlated with high correlation. A & B are constant terms and there are no exponential signs in the equation.
- In the **multiple regression model**, there are multiple x in the form of x1, x2, x3 and so on.
- **Intercept** – When we draw a line through the Y-axis and the distance between the regression line and Y axis is called the Intercept.
- **Slope of the Line** – It is the angle between the Regression line and the X-Axis is the Slope of the Line.

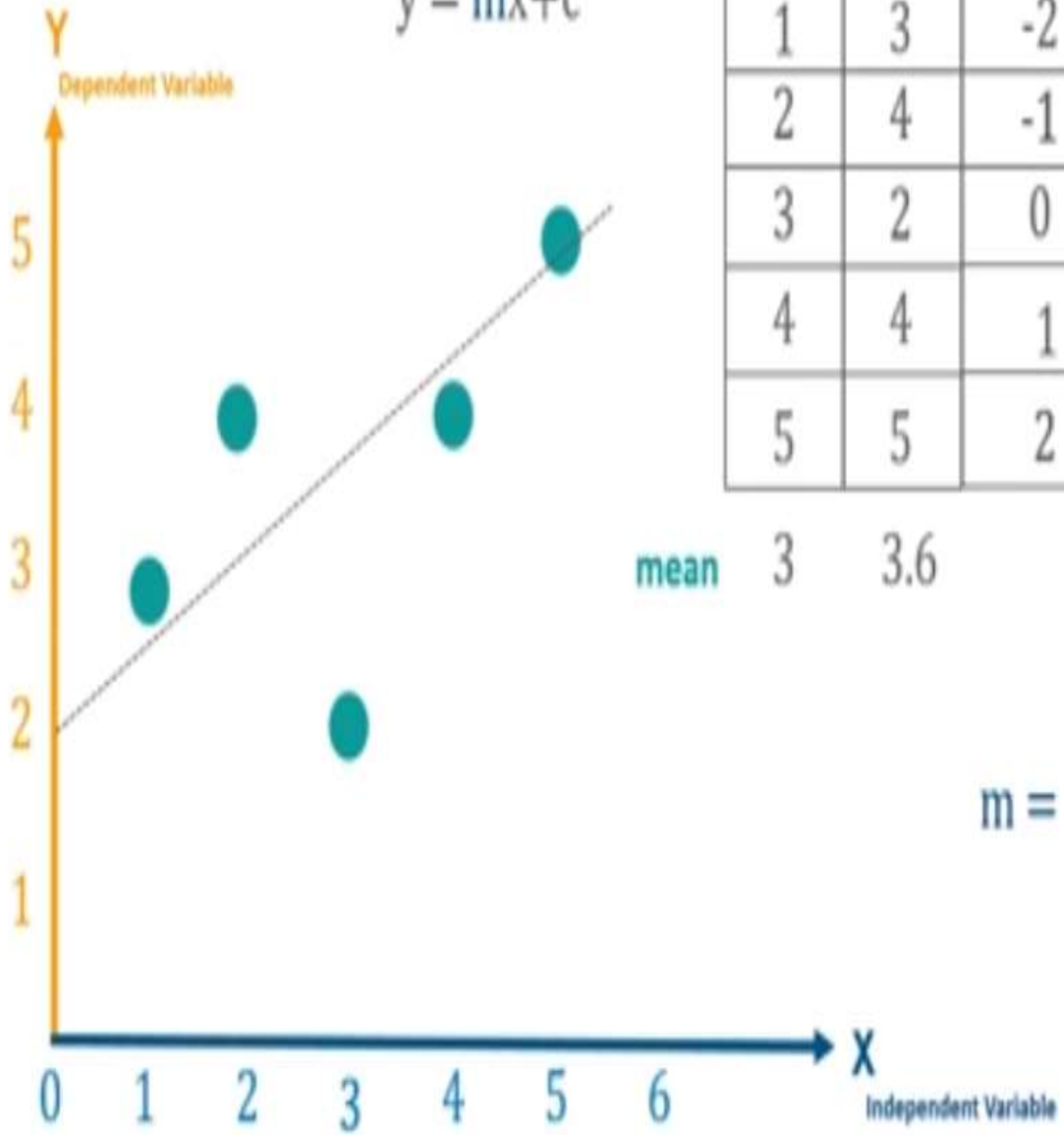
Find equation of line



x	y
1	3
2	4
3	2
4	4
5	5

$$m = \frac{\sum (x - \dot{x})(y - \dot{y})}{\sum (x - \dot{x})^2}$$

$y = mx + c$

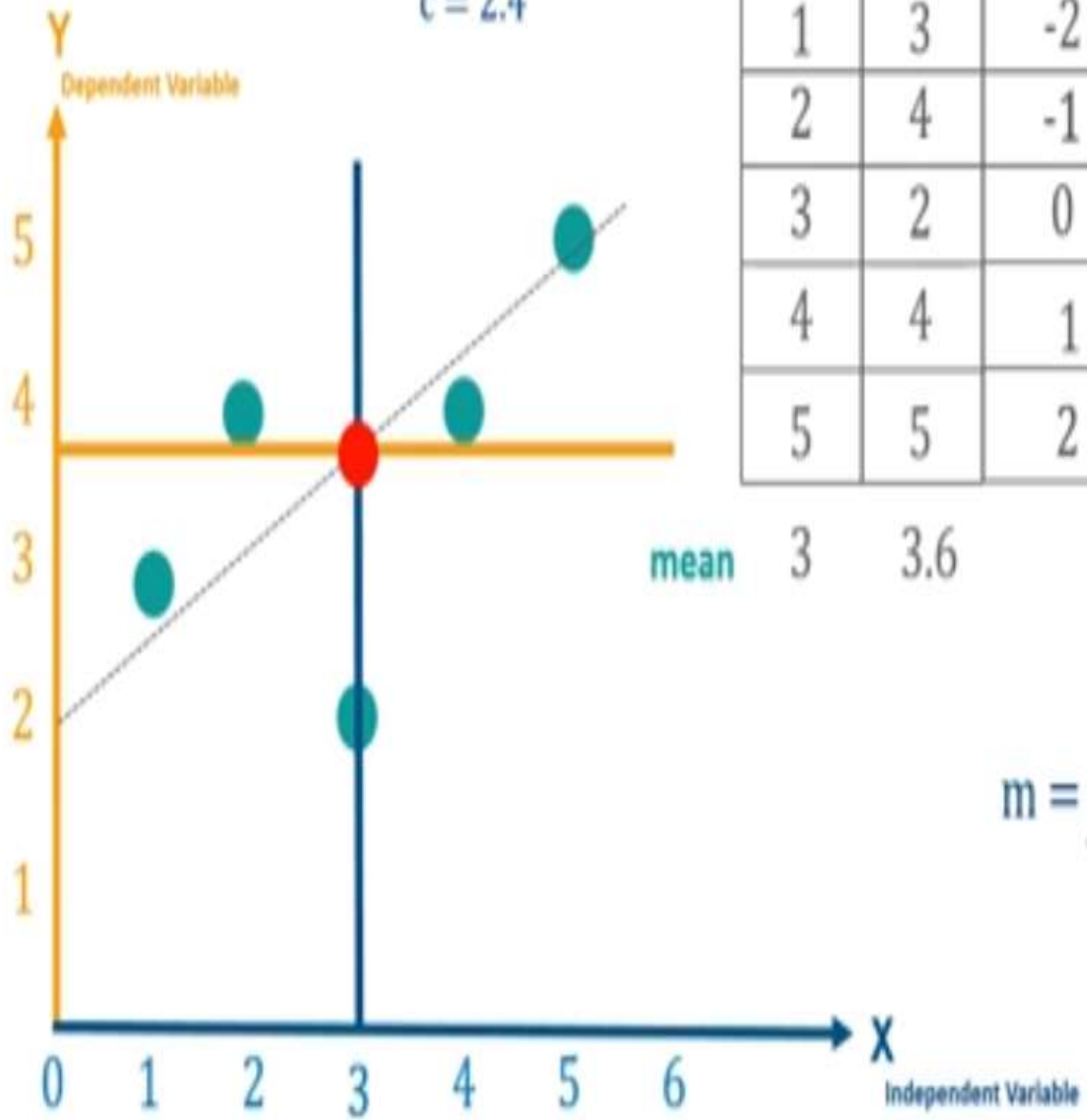


x	y	$x - \dot{x}$	$y - \dot{y}$	$(x - \dot{x})^2$	$(x - \dot{x})(y - \dot{y})$
1	3	-2	-0.6	4	1.2
2	4	-1	0.4	1	-0.4
3	2	0	-1.6	0	0
4	4	1	0.4	1	0.4
5	5	2	1.4	4	2.8
mean 3	3.6			$\Sigma = 10$	$\Sigma = 4$

$$m = \frac{\sum (x - \dot{x})(y - \dot{y})}{\sum (x - \dot{x})^2} = \frac{4}{10}$$

$y = mx + c$

$c = 2.4$



x	y	$x - \dot{x}$	$y - \dot{y}$	$(x - \dot{x})^2$	$(x - \dot{x})(y - \dot{y})$
1	3	-2	-0.6	4	1.2
2	4	-1	0.4	1	-0.4
3	2	0	-1.6	0	0
4	4	1	0.4	1	0.4
5	5	2	1.4	4	2.8
mean 3	3.6			$\Sigma = 10$	$\Sigma = 4$

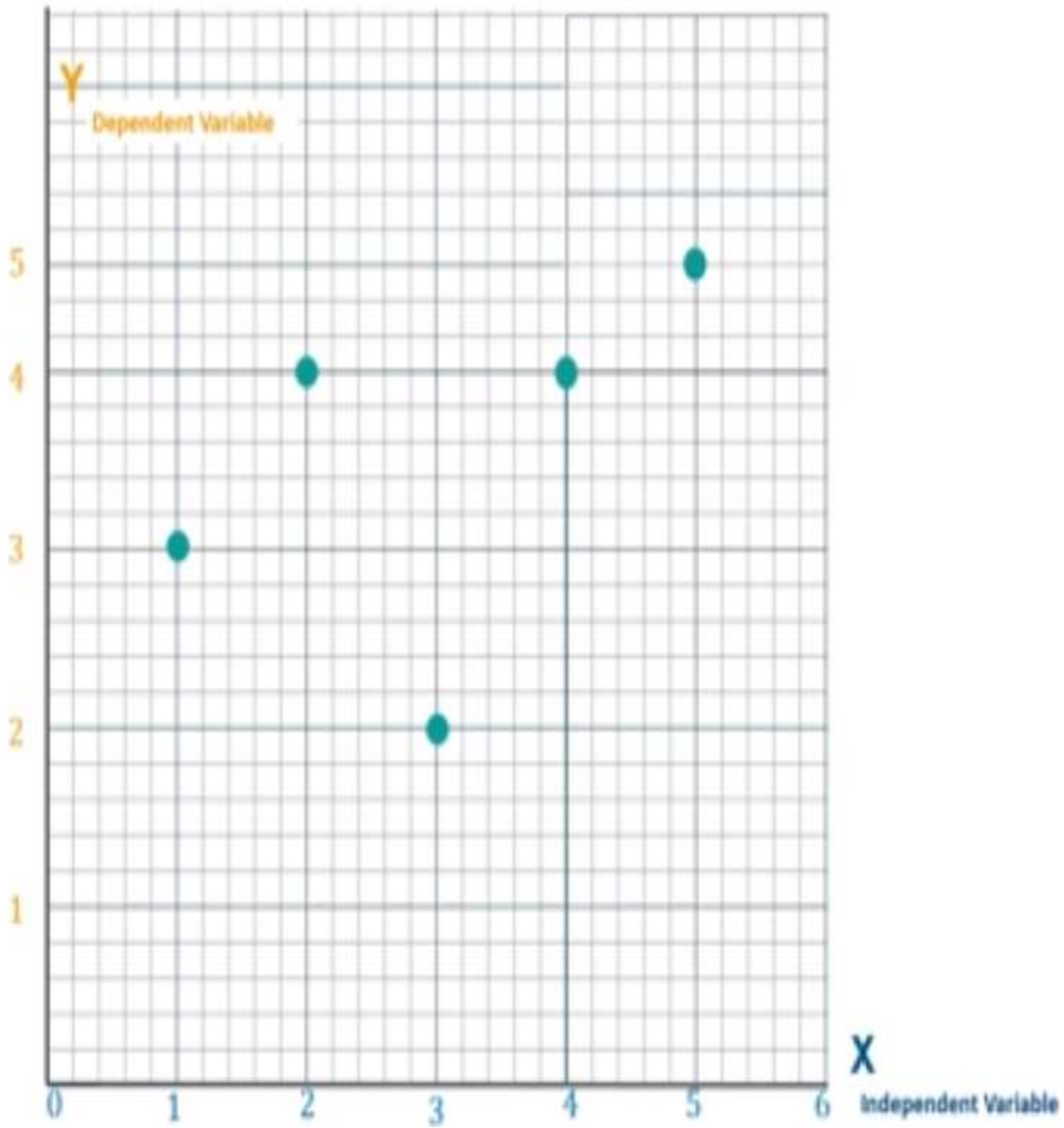
$$m = \sum \frac{(x - \dot{x})(y - \dot{y})}{(x - \dot{x})^2} = \frac{4}{10}$$

$m = 0.4$

$c = 2.4$

$y = 0.4x + 2.4$

Mean Square Error



$m = 0.4$

$c = 2.4$

$y = 0.4x + 2.4$

For given $m = 0.4$ & $c = 2.4$, lets predict values for y for $x = \{1,2,3,4,5\}$

$y = 0.4 \times 1 + 2.4 = 2.8$

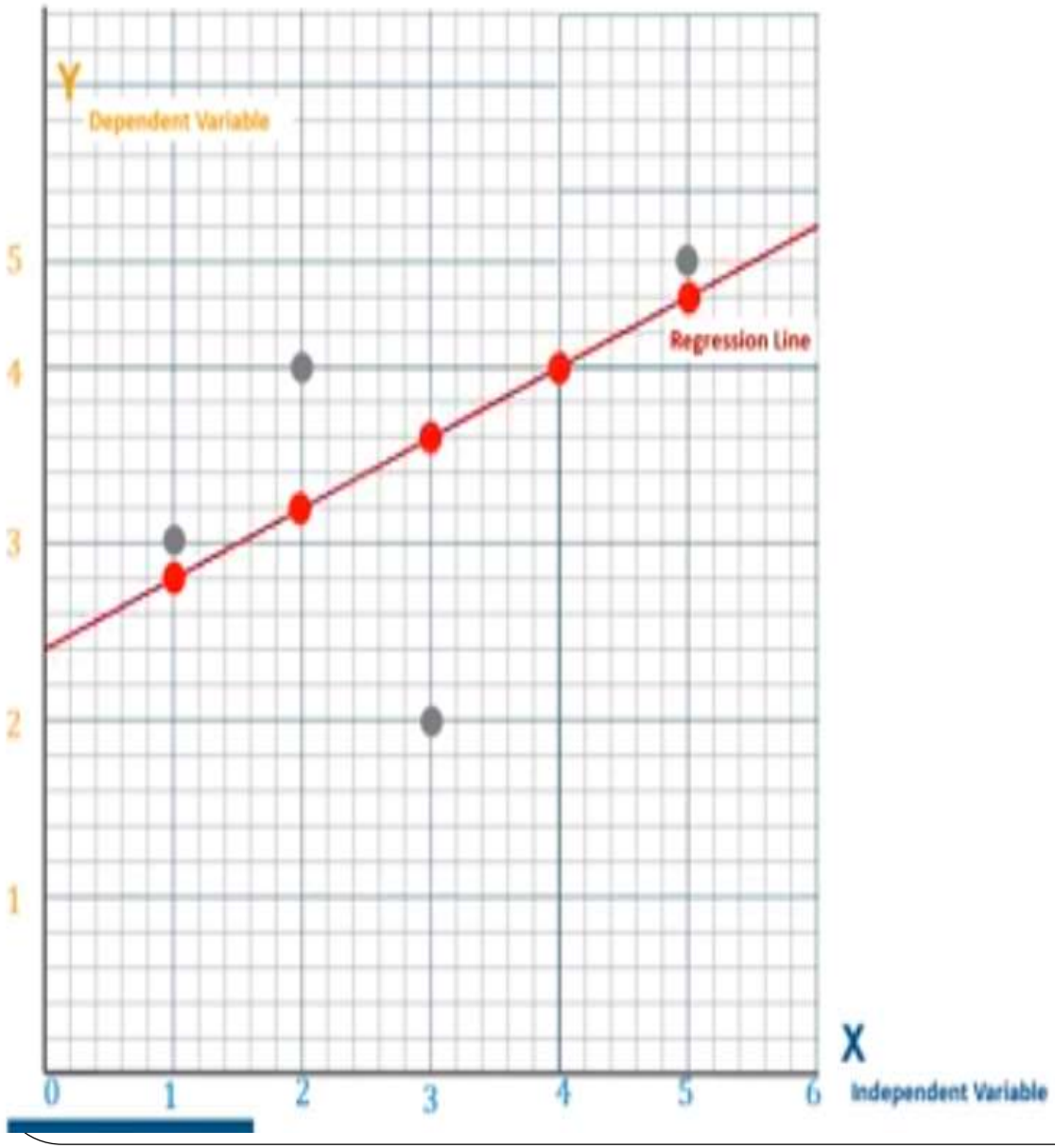
$y = 0.4 \times 2 + 2.4 = 3.2$

$y = 0.4 \times 3 + 2.4 = 3.6$

$y = 0.4 \times 4 + 2.4 = 4.0$

$y = 0.4 \times 5 + 2.4 = 4.4$

Mean Square Error



$m = 0.4$

$c = 2.4$

$y = 0.4x + 2.4$

For given $m = 0.4$ & $c = 2.4$, lets predict values for y for $x = \{1,2,3,4,5\}$

$y = 0.4 \times 1 + 2.4 = 2.8$

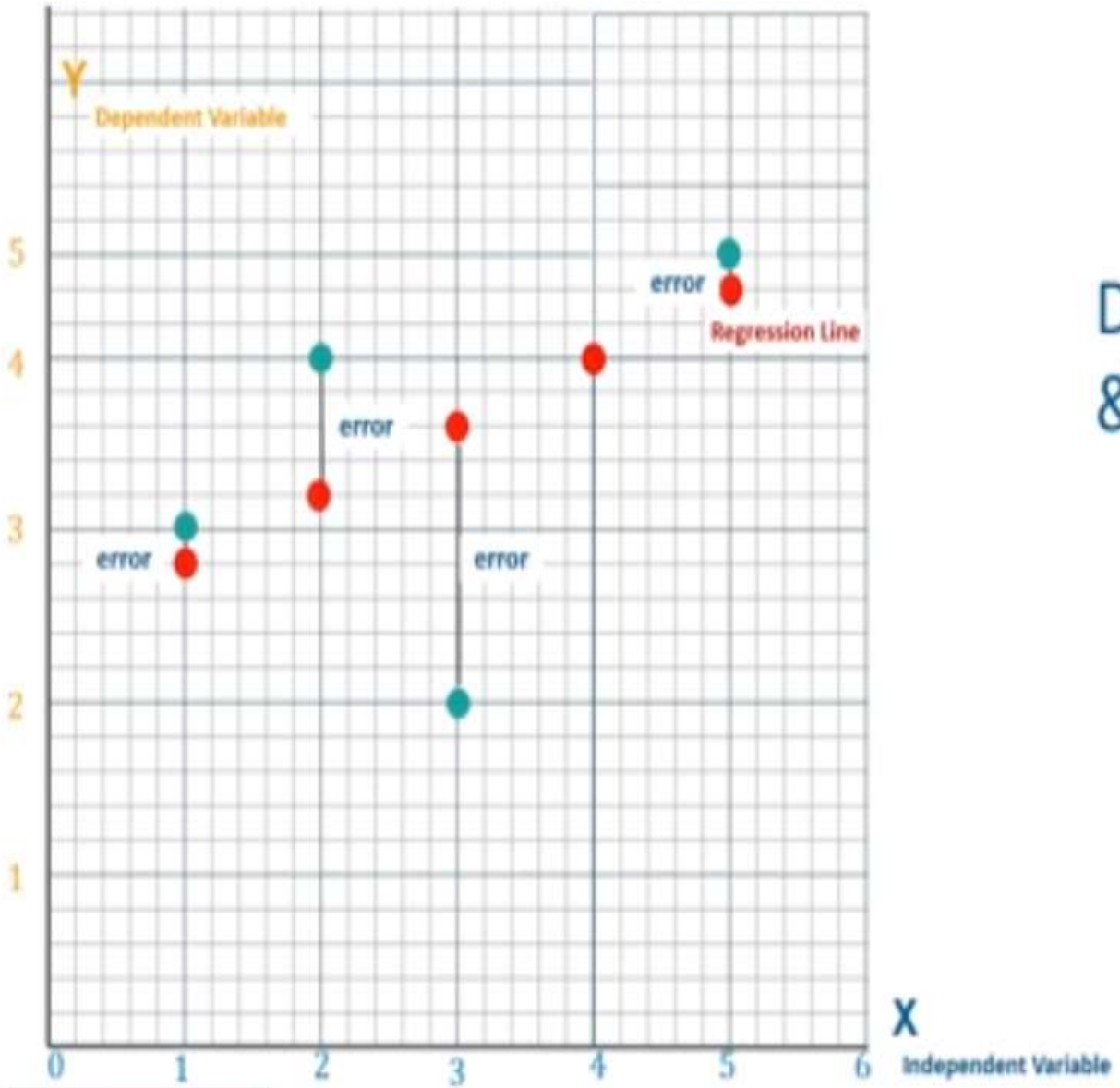
$y = 0.4 \times 2 + 2.4 = 3.2$

$y = 0.4 \times 3 + 2.4 = 3.6$

$y = 0.4 \times 4 + 2.4 = 4.0$

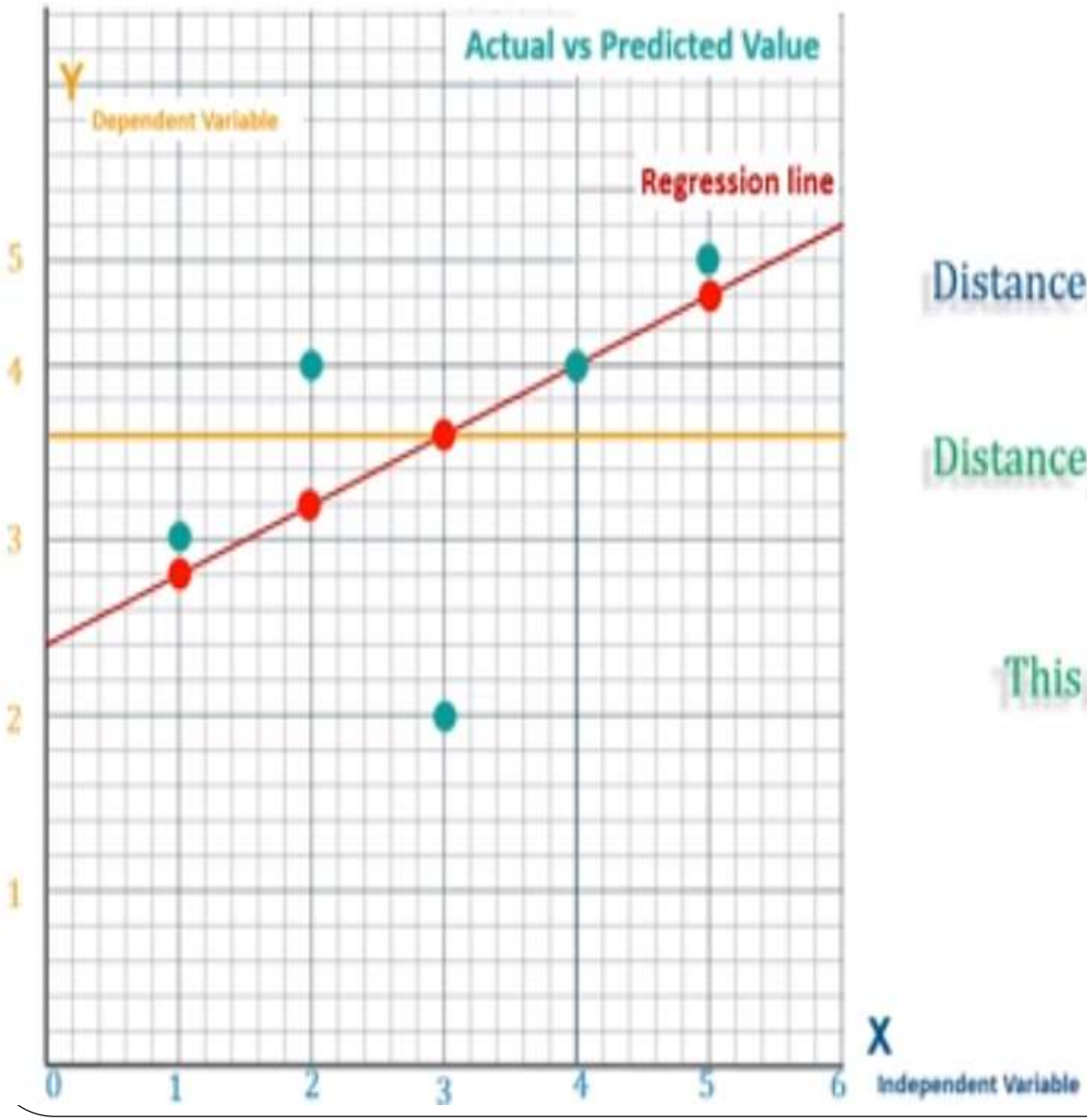
$y = 0.4 \times 5 + 2.4 = 4.4$

Mean Square Error



Distance between actual & predicted value

Calculation of R^2



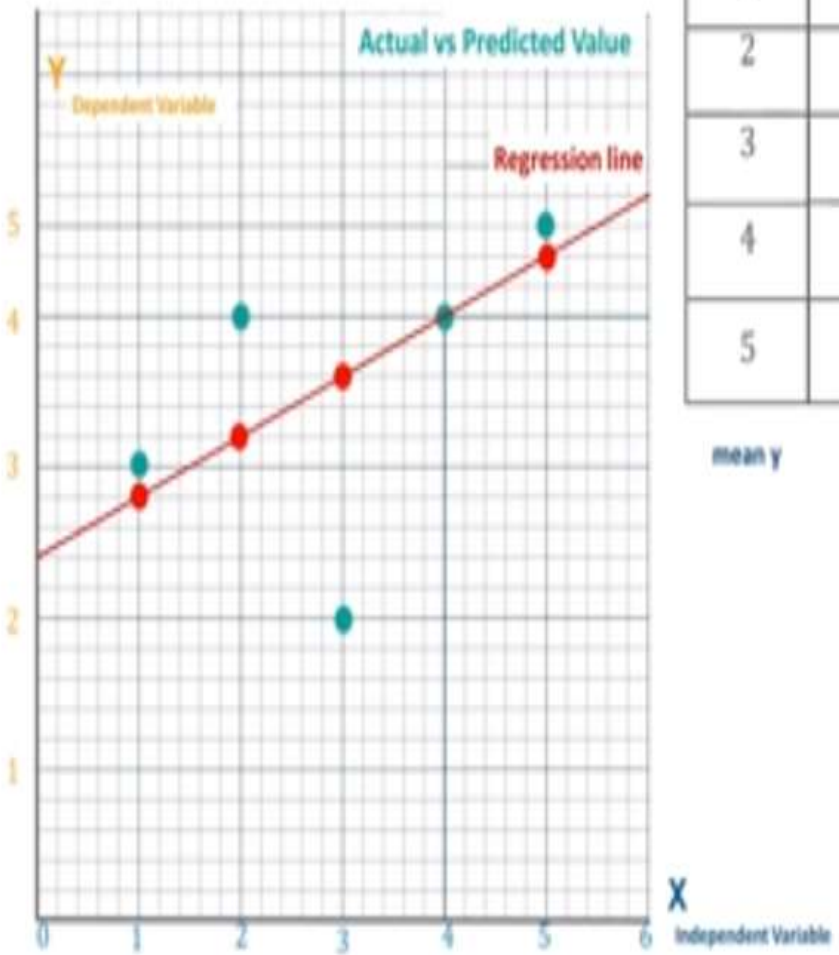
Distance actual - mean

vs

Distance predicted - mean

This is nothing but $R^2 = \dfrac{\sum (y_p - \bar{y})^2}{\sum (y - \bar{y})^2}$

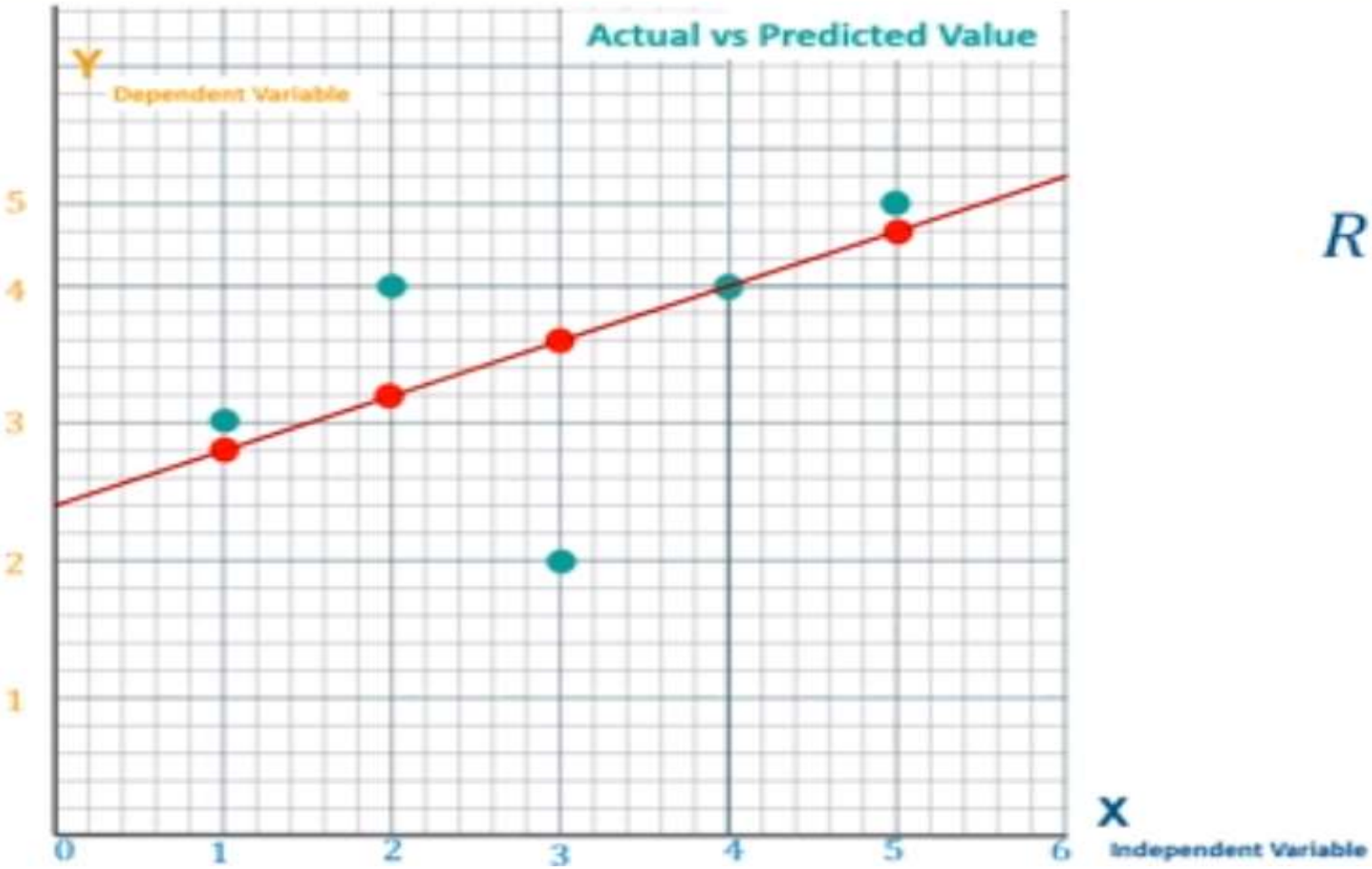
Calculation of R^2



x	y	$y - y$	$(y - y)^2$	y_p	$(y_p - y)$	$(y_p - y)^2$
1	3	-0.6	0.36	2.8	-0.8	0.64
2	4	0.4	0.16	3.2	-0.4	0.16
3	2	-1.6	2.56	3.6	0	0
4	4	0.4	0.16	4.0	0.4	0.16
5	5	1.4	1.96	4.4	0.8	0.64
mean y	3.6		$\sum$ 5.2			$\sum$ 1.6

$$R^2 = \frac{1.6}{5.2} = \frac{\sum \left((y_p - \bar{y})\right)^2}{\sum (y - \bar{y})^2}$$

Calculation of R^2



$R^2 \approx 0.3$

- R square is 0.3 means data points are far away from regression line.
- If R square is 1 than we say that actual line is on predicted line.

Advantages of linear regression

- Linear regression performs exceptionally well for linearly separable data.
- Easier to implement, interpret and efficient to train.
- It handles overfitting pretty well using dimensionally reduction techniques, regularization, and cross-validation.
- One more advantage is the extrapolation beyond a specific data set.

Disadvantages of linear regression

- The assumption of linearity between dependent and independent variables.
- It is often quite level to noise and overfitting.
- Linear regression is quite sensitive to outliers.

Simple Linear Regression

$$y = b_0 + b_1 x_1$$

Multiple Linear Regression

$$y = b_0 + b_1 x_1 + b_2 x_2 + \ldots + b_n x_n$$

Polynomial Linear Regression

$$y = b_0 + b_1 x_1 + b_2 x_1^2 + \ldots + b_n x_1^n$$

Multiple linear regression

- Multiple linear regression refers **to a statistical technique that is used to predict the outcome of a variable based on the value of two or more variables.**

- It is sometimes known **simply as multiple regression**, and it is **an extension of linear regression.**

- The variable that we want to predict is known as the dependent variable, while the variables we use to predict the value of the dependent variable are known as independent or explanatory (helpful) variables.

- Multiple Linear Regression is one of the important regression algorithms which models the linear relationship between **a single dependent continuous variable and more than one independent variable.**

For two independent variables, the general form of the multiple regression equation is:

$$\hat{Y} = a + b_1X_1 + b_2X_2$$

- X_1 and X_2 are the independent variables.
- a is the Y-intercept
- b_1 is the net change in Y for each unit change in X_1 holding X_2 constant. It is called a partial regression coefficient, a net regression coefficient, or just a regression coefficient.

- Multiple Linear Regression is one of the important regression algorithms which models the linear relationship between a single dependent continuous variable and more than one independent variable.

- For MLR, the **dependent or target variable(Y) must be the continuous/real, but the predictor or independent variable may be of continuous or categorical form.**

- Each feature variable must model the linear relationship with the dependent variable.

- MLR tries to fit a regression line through a **multidimensional space of data-points.**

Assumptions for Multiple Linear Regression

- A **linear relationship** should exist between the Target and predictor variables.
- The regression residuals must be **normally distributed**.
- MLR assumes little or **no multicollinearity** (correlation between the independent variable) in data.

Polynomial Regression

- Polynomial regression is a special case of linear regression where we fit a **polynomial equation on the data with a curvilinear relationship between the target variable and the independent variables**.

- Polynomial Regression is a regression algorithm that models the relationship between a dependent(y) and independent variable(x) as nth degree polynomial.

- The Polynomial Regression equation is given below:

  - $\mathbf{y= b_0+b_1x_1+ b_2x_1^2+ b_2x_1^3+..... b_nx_1^n}$

- It is also **called the special case of Multiple Linear Regression in ML**. Because we **add some polynomial terms to the Multiple Linear regression equation to convert it into Polynomial Regression.**

- It is a linear model with some modification in order **to increase the accuracy.**

- The dataset used in Polynomial regression **for training is of non-linear nature.**

- It makes use of a linear regression model to fit the complicated and non-linear functions and datasets.

- **Hence, "*In Polynomial regression, the original features are converted into Polynomial features of required degree (2,3,..,n) and then modeled using a linear model.*"**

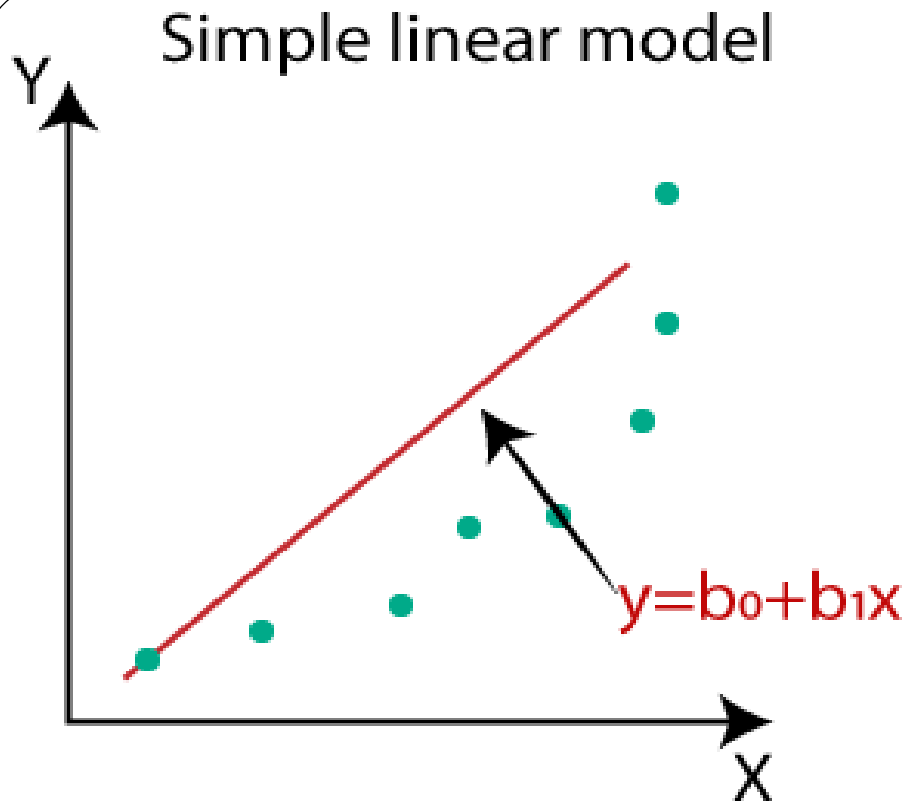
Need for Polynomial Regression

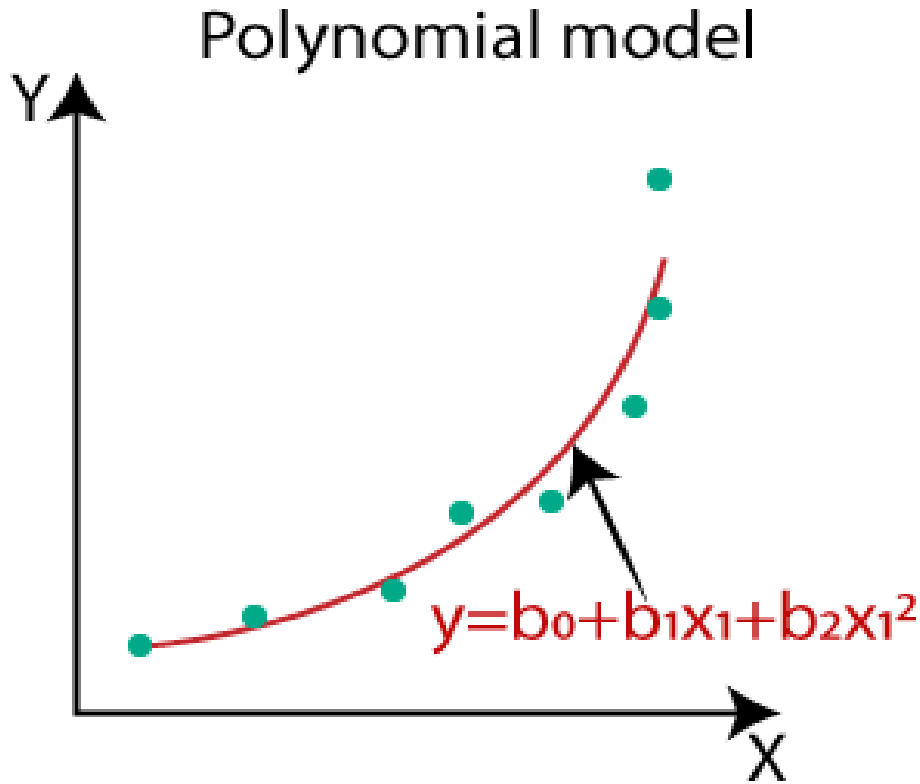
- If we apply a linear model on a **linear dataset**, then it provides us a good result as we have seen in Simple Linear Regression, but if we apply the same model without any modification on a **non-linear dataset**, then it will produce a drastic output.

- Due to which loss function will increase, the error rate will be high, and accuracy will be decreased.

- So for such cases, **where data points are arranged in a non-linear fashion, we need the Polynomial Regression model**.

- A Polynomial Regression algorithm is also called Polynomial Linear Regression because it does not depend on the variables, instead, it depends on the coefficients, which are arranged in a linear fashion.





•In the above image, we have taken a dataset which is arranged non-linearly. So if we try to cover it with a linear model, then we can clearly see that it hardly covers any data point.

•On the other hand**, a curve is suitable to cover most of the data points, which is of the Polynomial model.**

•Hence, *if the* ***datasets are arranged in a non-linear fashion****, then* ***we should use the Polynomial Regression model instead of Simple Linear Regression.***

- **Equation of the Polynomial Regression Model:**
- **Simple Linear Regression equation:**
  - $y = b_0+b_1x$ .....(a)
- **Multiple Linear Regression equation:**
  - $y= b_0+b_1x+ b_2x_2+ b_3x_3+....+ b_nx_n$ .....(b)
- **Polynomial Regression equation:**
  - $y= b_0+b_1x + b_2x^2+ b_3x^3+....+ b_nx^n$ .....(c)

- When we compare the above three equations, we can clearly see that all **three equations are Polynomial equations but differ by the degree of variables**.
- The Simple and Multiple Linear equations are also Polynomial equations with a single degree, and the Polynomial regression equation is Linear equation with the nth degree.
- So if **we add a degree to our linear equations, then it will be converted into Polynomial Linear equations.**

Logistic regression

- Logistic regression is one of the most popular Machine Learning algorithms, which comes under the Supervised Learning technique.
- It is used for **predicting the categorical dependent variable using a given set of independent variables.**
- Logistic regression **predicts the output of a categorical dependent variable**. Therefore the **outcome must be a categorical or discrete value**.
- It can be eitherYes or No, 0 or 1, true or False, etc. but instead of giving the exact value as 0 and 1, **it gives the probabilistic values which lie between 0 and 1**.
- Logistic Regression is much similar to the Linear Regression except that how they are used. **Linear Regression is used for solving Regression problems,**
- whereas **Logistic regression is used for solving the classification problems**.

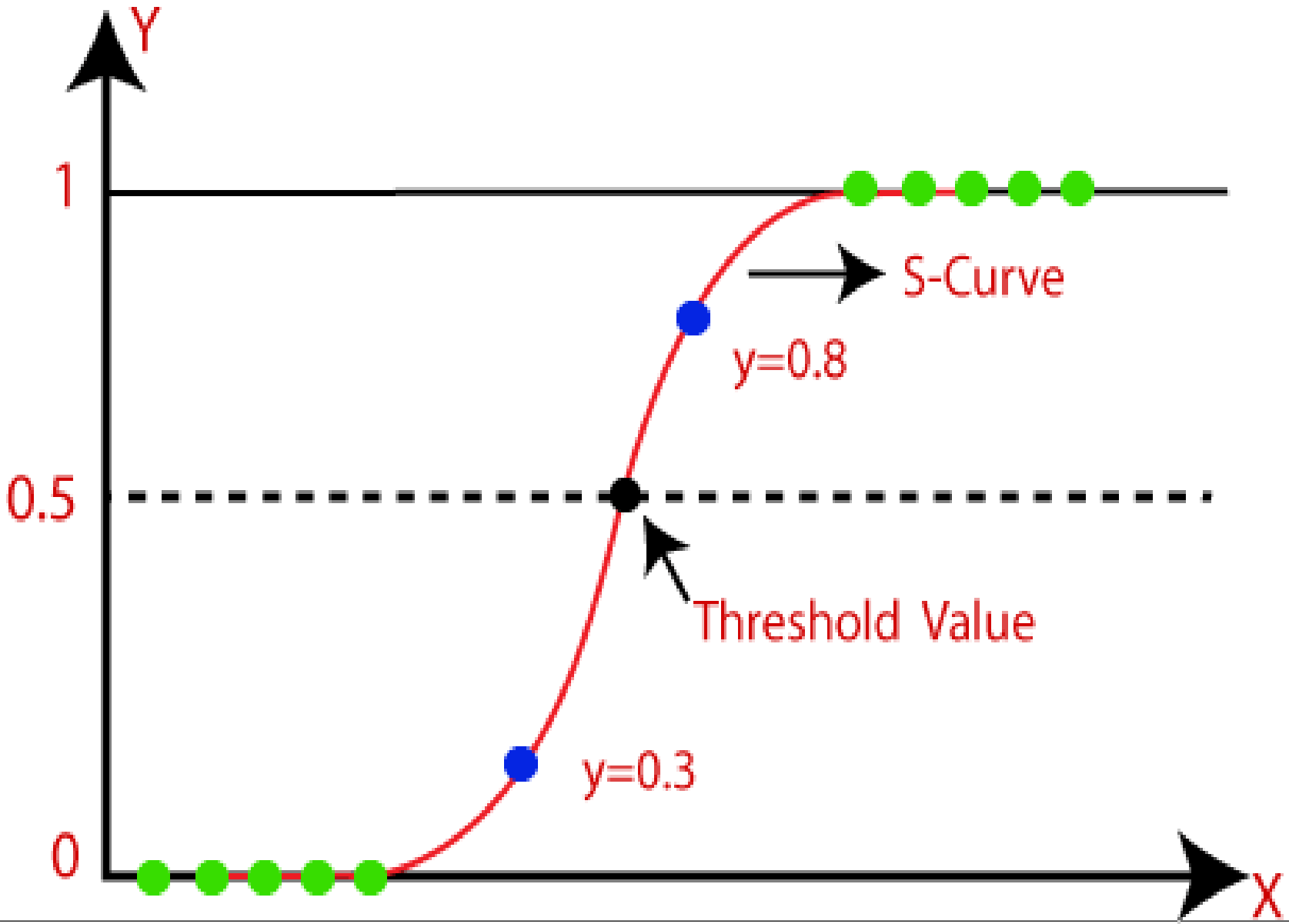
- In Logistic regression**, instead of fitting a regression line**, we **fit an "S" shaped logistic function, which predicts two maximum values (0 or 1).**

- The **curve from the logistic function indicates the probability of something such as whether the cells are cancerous or not, a mouse is fat or not based on its weight, etc.**

- Logistic Regression is a significant machine learning algorithm because it has the ability **to provide probabilities and classify new data using continuous and discrete datasets.**

- Logistic Regression can be used to classify the observations using different types of data and can **easily determine the most effective variables used for the classification.**

Logistic Regression



Logistic Function (Sigmoid Function):

- The sigmoid function is a mathematical function used to map the predicted values to probabilities.
- It maps any real value into another value within a range of 0 and 1.
- The value of the **logistic regression must be between 0 and 1, which cannot go beyond this limit, so it forms a curve like the "S" form.**
- **The S-form curve is called the Sigmoid function or the logistic function.**
- In logistic regression, we use the concept of the **threshold value,** which defines the probability of either 0 or 1. **Such as values above the threshold value tends to 1, and a value below the threshold values tends to 0.**

Assumptions for Logistic Regression:

- The dependent variable must be categorical in nature.

- The independent variable should not have multi-collinearity (correlation between the independent variable).

Logistic Regression Equation

The Logistic regression equation can be obtained from the Linear Regression equation. The mathematical steps to get Logistic Regression equations are given below:

- We know the equation of the straight line can be written as:

$$y = b_0 + b_1x_1 + b_2x_2 + b_3x_3 + \cdots + b_nx_n$$

- In Logistic Regression y can be between 0 and 1 only, so for this let's divide the above equation by (1-y):

$$\frac{y}{1-y}\text{; 0 for y= 0, and infinity for y=1}$$

- But we need range between -[infinity] to +[infinity], then take logarithm of the equation it will become:

$$log\left[\frac{y}{1-y}\right] = b_0 + b_1x_1 + b_2x_2 + b_3x_3 + \cdots + b_nx_n$$

Type of Logistic Regression

- On the basis of the categories, Logistic Regression can be classified into three types:
- **Binomial:** In binomial Logistic regression, there can be only two possible types of the dependent variables, such as 0 or 1, Pass or Fail, etc.
- **Multinomial:** In multinomial Logistic regression, there can be 3 or more possible unordered types of the dependent variable, such as "cat", "dogs", or "sheep“
- **Ordinal:** In ordinal Logistic regression, there can be 3 or more possible ordered types of dependent variables, such as "low", "Medium", or "High".

Linear Regression	Logistic Regression
Linear regression is used to predict the continuous dependent variable using a given set of independent variables.	Logistic Regression is used to predict the categorical dependent variable using a given set of independent variables.
Linear Regression is used for solving Regression problem.	Logistic regression is used for solving Classification problems.
In Linear regression, we predict the value of continuous variables.	In logistic Regression, we predict the values of categorical variables.
In linear regression, we find the best fit line, by which we can easily predict the output.	In Logistic Regression, we find the S-curve by which we can classify the samples.
Least square estimation method is used for estimation of accuracy.	Maximum likelihood estimation method is used for estimation of accuracy.
The output for Linear Regression must be a continuous value, such as price, age, etc.	The output of Logistic Regression must be a Categorical value such as 0 or 1, Yes or No, etc.
In Linear regression, it is required that relationship between dependent variable and independent variable must be linear.	In Logistic regression, it is not required to have the linear relationship between the dependent and independent variable.
In linear regression, there may be collinearity between the independent variables.	In logistic regression, there should not be collinearity between the independent variable.

Regression Algorithm	Classification Algorithm
In Regression, the output variable must be of continuous nature or real value.	In Classification, the output variable must be a discrete value.
The task of the regression algorithm is to map the input value (x) with the continuous output variable(y).	The task of the classification algorithm is to map the input value(x) with the discrete output variable(y).
Regression Algorithms are used with continuous data.	Classification Algorithms are used with discrete data.
In Regression, we try to find the best fit line, which can predict the output more accurately.	In Classification, we try to find the decision boundary, which can divide the dataset into different classes.
Regression algorithms can be used to solve the regression problems such as Weather Prediction, House price prediction, etc.	Classification Algorithms can be used to solve classification problems such as Identification of spam emails, Speech Recognition, Identification of cancer cells, etc.
The regression Algorithm can be further divided into Linear and Non-linear Regression.	The Classification algorithms can be divided into Binary Classifier and Multi-class Classifier.

Maximum likelihood estimation

The coefficients in a logistic regression are estimated using a process called Maximum Likelihood Estimation (MLE). First, let us understand what is likelihood function before moving to MLE. A fair coin outcome flips equally heads and tails of the same number of times. If we toss the coin 10 times, it is expected that we get five times Head and five times Tail.

Let us now discuss about the probability of getting only Head as an outcome; it is $5/10 = 0.5$ in the above case. Whenever this number (P) is greater than 0.5, it is said to be in favour of Head. Whenever P is lesser than 0.5, it is said to be against the outcome of getting Head.

Let us represent 'n' flips of coin as $X_1, X_2, X_3, \ldots, X_n$. Now X_i can take the value of 1 or 0.

$X_i = 1$ if Head is the outcome

$X_i = 0$ if Tail is the outcome

When we use the Bernoulli distribution represents each flip of the coin:

$$f(x_i|\theta) = \theta^{x_i}(1 - \theta)^{1-x_i}$$

Each observation X_i is independent and also identically distributed (iid), and the joint distribution simplifies to a product of distributions.

$$f(x_1, \ldots, x_n|\theta)_{i=1}^{n} f(x_i|\theta) = \theta^{x_1}(1 - \theta)^{1-x_1} \ldots \theta^{x_n}(1 - \theta)^{1-x_n} = \theta^{\#H}(1 - \theta)^{n-\#H},$$

where $\#H$ is the number of flips that resulted in the expected outcome (heads in this case).

The likelihood equation is

$$L(\theta|x) = {}_{i=1}^{n} f(x_i|\theta)$$

But the likelihood function is not a probability. The likelihood for some coins may be 0.25 or 0 or 1.

MLE is about predicting the value for the parameters that maximizes the likelihood function.

$$\log L(\theta|x) = \sum_{i=1}^{n} \log f(x_i|\theta)$$